Rules of Origin and the Use of NAFTA

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Abstract

We study how RoO affect the use of NAFTA. To qualify for preferential tariffs, exporters using NAFTA must comply with RoO. We document: (i) the smallest and largest firms use NAFTA less intensively than medium-sized firms, and (ii) the reduction in sourcing from non-NAFTA countries when using NAFTA increases with firm size. We rationalize these findings in a partial equilibrium model of input sourcing with fixed costs of using NAFTA and of sourcing from foreign suppliers, where the opportunity cost of complying with RoO rises with firm size. Using data on Mexican exporters, we quantify the model and conduct counterfactuals. A 20 p.p. increase in RoO strictness lowers U.S. intermediate exports to Mexico by 3.89%, while a 20 p.p. tariff increase raises them by 3.18%. Additional exercises highlight the importance of accounting for adjustment costs, and that although stricter RoO can increase regional content, they reduce the scale of trade.

Keywords: *NAFTA, Rules of Origin, MFN tariffs, Trade Policy, Input Sourcing*

JEL Codes: F12, F13, F15, F23, O19

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1 Introduction

Since the creation of the World Trade Organization (WTO) in 1995, Free Trade Agreements (FTAs) have proliferated worldwide. Bhagwati (1995) refers to this as the *Spaghetti Bowl Effect*, describing how regional agreements increasingly dominate trade over multilateral organizations such as the WTO.[1] A common feature of FTAs is Rules of Origin (RoO), defined as *"the criteria needed to determine the national source of a product."*[2] RoO are pervasive, included in at least 62% of existing FTAs according to Kniahin and De Melo (2022). RoO aim to protect local industries and increase regional content in exports by requiring firms to source part of their intermediates from suppliers in member countries. This creates a tradeoff between potential efficiency losses and lower trade barriers. Compliance prevents firms from sourcing inputs from the most efficient global suppliers, raising production costs, but allows them to benefit from preferential tariffs, lowering export costs to member countries.

This paper examines the effect of RoO on the use of the North American Free Trade Agreement (NAFTA) and their implications for bilateral trade flows. We use a novel dataset covering the universe of Mexican exporters, identifying whether firms use NAFTA , i.e., benefit from preferential tariffs by complying with RoO, or not for their exports to the US, as well as the foreign inputs they source. We complement this with: (i) the Most Favored Nation (MFN) tariffs the US applies to Mexican exporters not using NAFTA, (ii) the RoO required for NAFTA use, and (iii) the input composition of each product exported to the US. We construct a value-based measure of RoO strictness, defined as the share of a product's input value that must be sourced from NAFTA suppliers. For this, we leverage the fact that most NAFTA RoO are specified as *Classification Changes*, as documented by Conconi et al. (2018), which stipulate for each product the inputs that must come from NAFTA countries based on the Harmonized System (HS) codes of both the final product and its intermediates.[3]

We document three empirical facts. First, RoO strictness is positively correlated with MFN tariffs. Second, NAFTA use and firm size follow an inverse U-shape: medium-sized exporters are more likely than smaller or larger ones to use NAFTA. Third, the share of firms sourcing from non-NAFTA suppliers is lower among NAFTA exporters, with the gap widening across exporter size deciles. These patterns hold when controlling for industry fixed effects, capturing sectoral factors affecting NAFTA use and foreign sourcing. The first fact implies a tradeoff: avoiding high tariffs requires sourcing a large share of inputs from NAFTA suppliers. The second and third facts suggest that Mexican exporters face fixed

[1] As of 2025, there are 375 active FTAs worldwide. Source: WTO's RTA-IS.

[2] Source: WTO's Technical Information on Rules of Origin.

[3] For example, a firm exporting a product with HS code 85 must source all inputs with HS code 85 from NAFTA countries, while inputs with other HS codes can be freely sourced.

costs both in using NAFTA —such as compliance with labor rules and learning to use the FTA —and in sourcing from foreign countries, including search and bureaucratic costs.

We incorporate these fixed costs into the partial equilibrium model of global input sourcing by Antras et al. (2017), extending it to multiple sectors and industries. Mexican firms export to the US and choose: (i) whether to use NAFTA or WTO to export, and (ii) the set of countries from which to source inputs. Using NAFTA requires compliance with RoO and payment of the associated fixed cost, while using WTO entails paying MFN tariffs. Firms are monopolistically competitive and require a continuum of intermediate inputs sourced from heterogeneous suppliers worldwide. RoO are modeled as restricting a share of inputs to NAFTA suppliers, with the rest sourced from the most efficient suppliers among countries for which fixed sourcing costs have been paid. Mexican exporters differ in the RoO and MFN tariffs they face, their fixed costs of using NAFTA and sourcing abroad, the attractiveness of foreign sourcing, and their export-specific productivity.

The model predicts how NAFTA usage responds to RoO, tariffs, sourcing efficiency, and fixed costs: (i) it decreases with RoO, which raise marginal costs by forcing sourcing from potentially inefficient suppliers; (ii) it increases with higher tariffs, which raise the cost of exporting under WTO; (iii) it increases with the efficiency of NAFTA suppliers, which lowers the opportunity cost of complying with RoO; (iv) it decreases with the efficiency of non-NAFTA suppliers; and (v) it decreases with NAFTA's fixed cost but increases with fixed costs of sourcing from foreign countries, since higher sourcing costs reduce the likelihood of accessing those suppliers, lowering the cost of RoO compliance. In the model, stricter RoO reduce bilateral trade in final goods, as sourcing from inefficient suppliers raises prices and lowers demand, and have ambiguous effects on bilateral trade in intermediates. Stricter RoO raise the share of inputs sourced from member countries, but lower final goods trade also reduces intermediate purchases. The net effect depends on the relative efficiency of NAFTA suppliers and the extent to which firms can source from non-NAFTA countries.

To quantify these tradeoffs, we estimate the model at the sectoral level, recovering fixed costs of using NAFTA and sourcing from foreign countries, as well as the US market demand faced my Mexican exporters. We target: (i) the share of firms using NAFTA, which identifies its fixed cost; (ii) the share sourcing from each foreign country, which identifies sourcing fixed costs; and (iii) median firm exports, which pins down US market demand by sector. The model is fed with product-level RoO and tariff data. We also estimate how attractive foreign input sourcing is for each Mexican industry.

With the model calibrated, we run counterfactuals to quantify the effects of higher RoO strictness and MFN tariffs, highlighting the distinct channels through which each policy shapes trade and firm behavior. In the first counterfactual, a uniform 20 percentage-point increase in RoO strictness[4] reduces US-CA intermediate exports to Mexico by about 3.9%, along with declines in Mexican exports and profits. Stricter rules force NAFTA users to replace efficient non-NAFTA suppliers with less efficient regional ones, raising marginal costs, lowering output, and reducing demand for all inputs, including regional ones. Larger firms are hit harder, as their broader sourcing networks include more non-NAFTA suppliers, making the efficiency loss outweigh tariff savings and discouraging NAFTA use. In contrast, a 20 percentage-point increase in MFN tariffs increases US-CA intermediate exports to Mexico, as many Mexican firms shift into NAFTA to avoid higher tariffs. Regional sourcing becomes more attractive, prompting purchases to move from non-NAFTA to US-CA suppliers. While this boosts intermediate imports from the region, efficiency losses arise when firms replace more efficient non-NAFTA inputs with costlier NAFTA ones. Higher marginal costs then reduce Mexican exports and profits, even as US-CA intermediate exports grow.

Two factors explain the heterogeneity in the responses to these scenarios. First, the distribution of firm size: fixed costs of foreign sourcing make the opportunity cost of complying with RoO rise with firm size, so protectionist policies are costlier in sectors with larger firms. Second, the relative efficiency of NAFTA versus non-NAFTA suppliers: sectors sourcing inputs where NAFTA is competitive are less affected by stricter RoO, while those relying on inputs where non-NAFTA suppliers have an advantage face greater efficiency losses and trade declines. Adjustment frictions play a key role in shaping short-run and heterogeneous responses to policy changes. If firms cannot quickly alter sourcing —due to contracts, search costs, or supplier certification —NAFTA use may rise simply because firms remain locked into arrangements that minimize immediate tariff costs. These frictions can vary by firm size and sector, with larger firms or those relying on specialized intermediates facing greater barriers. Across industries, stricter RoO can raise the share of regional content but consistently reduce the total value of regional trade, as replacing more efficient non-NAFTA suppliers increases costs and lowers output. While RoO were originally intended to prevent trade diversion, these results question their current aim of maximizing intra-regional trade, as higher regional content may come at the expense of overall trade volume.

This paper relates to several strands of literature, particularly on trade liberalization and Free Trade Agreements, including Trefler (2004), Romalis (2007), Arkolakis et al. (2008), Bustos (2011), Ossa (2011), Antràs and Staiger (2012), Kehoe and Ruhl (2013), Caliendo and Parro (2015), De Loecker et al. (2016), among others. To our knowledge, it is the first to provide evidence on which firms use NAFTA to export and how this choice depends on characteristics such as firm size and industry-specific factors.

Second, our work builds on the literature on firm-level global sourcing decisions, including

[4]For example, a product with 10% RoO strictness would rise to 30%.

Antras and Helpman (2004), Goldberg et al. (2010), Rodríguez-Clare (2010), Garetto (2013), Kee and Tang (2016), Tintelnot (2017), Bernard et al. (2018), and Head and Mayer (2019). Our framework extends Antras et al. (2017), which models firms' choice of sourcing countries, by adding a new decision margin: whether to export under NAFTA or WTO, capturing the tradeoff between lower tariffs and potential production inefficiencies.

Third, we contribute to the literature on content protection and the effects of RoO on firm behavior, beginning with Dixit and Grossman (1982), Grossman and Helpman (1992), Krishna and Krueger (1995), and Estevadeordal (1999), and including more recent work by Carrère and De Melo (2004), Anson et al. (2005), Augier et al. (2005), Ju and Krishna (2005), Cadot et al. (2006), Deardorff (2018), Krishna et al. (2021), Acosta and Leal (2022), and Ornelas and Turner (2023). Closely related is Conconi et al. (2018), who show that more restricted inputs under RoO saw greater substitution from non-NAFTA to NAFTA sourcing, and Head et al. (2024), who model RoO and input sourcing in the North American auto industry, identifying a Laffer Curve between RoO and regional content. We extend this literature by using comprehensive Mexican export transaction data to document new facts on NAFTA use and structurally estimate a model of RoO compliance.

The rest of the paper is organized as follows. Section 2 describes the data and how we combine sources to capture firm behavior and the costs and benefits of using NAFTA. Section 3 presents three empirical facts that guide our model and provide evidence of fixed costs of using NAFTA and sourcing from foreign countries. Section 4 develops the model, Section 5 describes its quantification, and Section 6 presents counterfactual scenarios. Section 7 concludes with main findings and policy implications.

2 Data

This section describes the data sources used to build a unique dataset on NAFTA use and firms' sourcing choices, including the RoO and tariffs faced by exporters to the US. We study Mexican firms exporting final products to the US under NAFTA or WTO and importing intermediate inputs globally. The data come from the following sources:

Mexican Customs Data: Accessed through the EconLab at Banco de México, this dataset covers the universe of export and import transactions. For each transaction, we observe the firm ID, HS 6-digit product, trade value, origin/destination, and whether it involves an intermediate or final good. Critically, we also know whether a transaction used NAFTA or WTO membership.

US 2022 Harmonized Tariff Schedule: Provides the HS 6-digit MFN tariffs that Mexican exporters would pay if exporting under WTO membership.

NAFTA's RoO: From Annex 401 of the NAFTA agreement, compiled by Conconi et al. (2018), this dataset gives the HS 6-digit RoO that Mexican firms had to meet to export under NAFTA. Most are defined as *Classification Changes*, specifying which inputs for each final product had to be sourced from suppliers in NAFTA countries.

BEA's 2017 Input-Output Tables: We use the Direct Requirement Coefficients Tables from the BEA, which provide us with a survey-based estimate of the average input composition of each final product. Products follow the BEA classification, similar to NAICS, allowing reclassification into the Harmonized System.

We combine the last two sources to construct a value-based measure of RoO strictness, which we define as the share of a product's total input value that must be sourced from NAFTA countries for the product to qualify for preferential treatment under NAFTA. The calculation uses two key pieces of information: (i) the input-value composition of each final product, as given by the BEA's Direct Requirement Coefficients (DRC) tables, and (ii) the set of inputs restricted to NAFTA sourcing according to Annex 401 of the NAFTA agreement. First, we identify, for each HS 6-digit product, the relevant inputs subject to sourcing restrictions based on their HS classification. Next, we weight these restricted inputs by their relative contribution to the product's total input value, as reported in the DRC tables, to determine the fraction of input value that must be sourced regionally. The resulting measure captures the intensity of the sourcing restriction at the product level: higher values indicate that a greater share of a product's input value must be sourced from NAFTA suppliers to comply with RoO. This process requires reclassifying the BEA's DRC tables from their original BEA product classification into the Harmonized System, ensuring full alignment between input composition data and the HS-based RoO requirements.[5]

We examine exports from September 2014, when NAFTA usage data first became available, to June 2020, the last month before the transition to USMCA. Over this period, 25,572 firms exported 1,050 unique products to the US under either NAFTA or WTO, yielding 1,019,408 firm-time-product observations. We then apply two sample restrictions. First, we drop firms whose total exports to all destinations are lower than their total imports, removing 29.27% of firms. Since the goal is studying the behavior of Mexican exporters, this excludes firms that are more likely to act as trade intermediaries or firms primarily serving the domestic market, which tend to have disproportionately large imports.[6]

[5]Details on the reclassification procedure are provided in Appendix A.
[6]For these firms the median imports-to-exports ratio is 254.84.

Second, at the product level, we drop HS 6-digit codes where: (i) tariffs are zero or not expressed in ad-valorem terms,[7] or (ii) exporters can opt to comply with an alternative Value-Added rule instead of RoO. The data in Conconi et al. (2018) includes information on the HS 6-digit codes for which this is the case. Products with positive ad-valorem tariffs account for 63.27% of all products, and 86.76% require compliance with RoO. After these restrictions, the final sample covers 410 HS 6-digit products in 48 HS 2-digit industries and 9,918 firms exporting final goods to the US under either NAFTA or WTO.

An illustrative example of an observation in our data is the following: In our customs data we observe that *Firm A* exported *Product 1* to the US using NAFTA, imported *Input 1* from the US, and *Input 2* from China. From the DRC Tables we know *Product 1* is made of 40% *Input 1*, 40% *Input 2*, and 20% *Input 3*. From RoO data, we know that if a firm uses NAFTA to export *Product 1*, it must source *Input 1* and *Input 3* from NAFTA countries, implying a RoO strictness of 60%. Lastly, MFN tariff data indicates that exporting *Product 1* under WTO requires paying a 10% ad-valorem tariff. Appendix B reports several descriptive tables and figures regarding the RoO strictness and MFN tariffs for the products in our sample.

3 Three Empirical Facts

This section documents three empirical facts on how RoO affect firm behavior and how they are rationalized in our model. First, RoO strictness and MFN tariffs are positively correlated. Second, small and large firms use NAFTA less than medium-sized firms. Third, the distortion RoO impose on firms' input sourcing grows with firm size.

Empirical Fact 1. *There is a positive correlation between RoO strictness and MFN tariffs.*

Figure 1 shows this positive correlation at the product level.[8] This implies firms face a tradeoff when choosing between NAFTA and WTO: products with higher MFN tariffs also have stricter RoO. In our model, firms weigh the cost of complying with RoO against the savings from avoiding MFN tariffs.

Empirical Fact 2. *The relationship between the use of NAFTA and firm size follows an inverse U-shape.*

Figure 2 shows NAFTA usage by firm size, proxied by average monthly exports to all destinations. Firms are grouped into size percentiles, and for each, we compute the share exporting under NAFTA. Usage is lower for small and large firms than for medium-sized

[7] For example, *US dollars per pound*, for which it is difficult to compare across products.
[8] This is a binscatter, grouping our 410 HS 6-digit products into 50 MFN tariff bins.

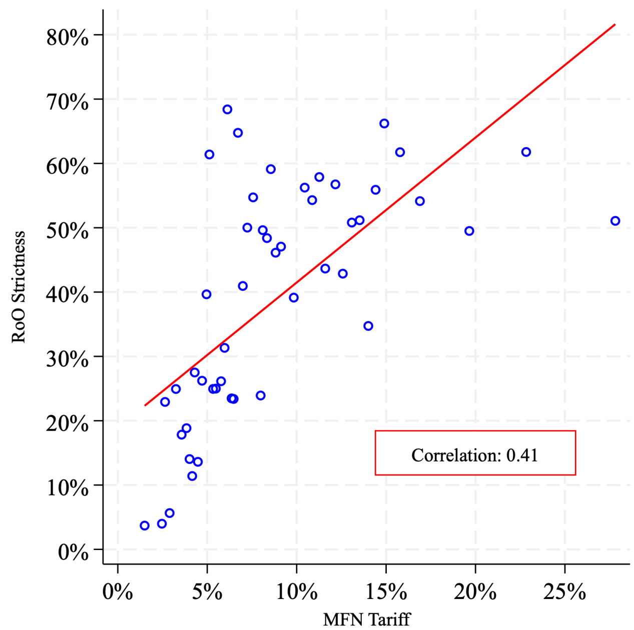


Figure 1: Correlation between RoO strictness and MFN tariffs.

ones. This pattern could reflect selection into industries rather than size itself, if small and large firms operate in sectors with weaker incentives to use NAFTA. To test this, we estimate Equation (1) using OLS.

$$\mathbb{N}_{ikjt} = \beta_0 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \alpha_1 \text{RoO}_j + \alpha_2 \text{MFN}_j + \iota_t + \epsilon_{ikjt} \tag{1}$$

where $\mathbb{N}_{ikjt} = 1$ if firm i in size decile k exports product j at time t using NAFTA, and $\mathbb{I}_{ikt} = 1$ if the firm is in size decile k. We control for RoO_j, the share of restricted inputs for product j under NAFTA, and MFN_j, its ad-valorem tariff under WTO, to capture the role of RoO and tariffs in NAFTA use. Time fixed effects ι_t control for common shocks. We omit industry fixed effects to retain the *ceteris paribus* effects of RoO and tariffs, as within-industry variation in these measures is limited.[9]

Figure 3 plots the estimated intercepts $\hat{\beta}_0 + \hat{\beta}_k$ across size deciles k, with the x-axis showing size deciles and the y-axis the predicted share of firms using NAFTA. We set $\text{RoO}_j = 0$ and $\text{MFN}_j = 0$ to isolate the direct effect of firm size. Results indicate that firm size partly explains the inverse U-shape in NAFTA use, even after controlling for RoO and tariffs. Estimates for $\hat{\alpha}_1$ and $\hat{\alpha}_2$ imply that a 1 s.d. increase in RoO strictness or MFN tariffs changes the probability of NAFTA use by -9.03 and 3.66 p.p., respectively.

[9]The average coefficient of variation in RoO strictness and MFN tariffs across products within an industry is 0.63 and 0.19, respectively.

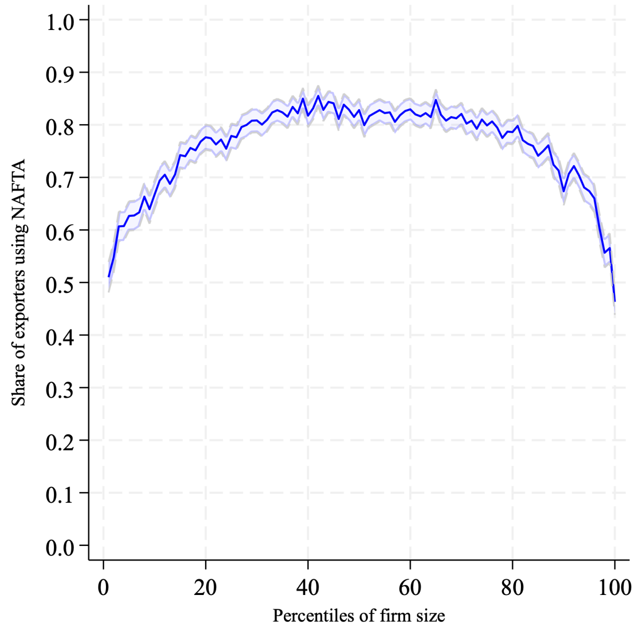


Figure 2: Share of exporters using NAFTA by size percentile.

Table C.1 in Appendix C reports the estimates and presents robustness checks: (i) adding industry fixed effects, (ii) using alternative specifications and size proxies, and (iii) verifying that results are not driven by any single industry.

Empirical Fact 3. *The decrease in the probability of sourcing inputs from non-NAFTA countries when using NAFTA to export is increasing in firm size.*

To study the impact RoO have on firms' input sourcing choices, we use OLS to estimate Equation (2) separately for firms using either NAFTA or WTO membership to export:

$$\mathbb{S}_{ikst} = \beta_0 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \iota_{s,t} + \epsilon_{ijt} \tag{2}$$

where $\mathbb{S}_{ikst} = 1$ if firm i of size k from industry s at time t sources inputs from outside NAFTA, i.e., imports intermediates from countries other than the US or Canada, and $\mathbb{I}_{ikt} = 1$ if the firm belongs to size decile k. We include industry-year fixed effects ι_{st} to capture unobservable heterogeneity in the attractiveness of sourcing from non-NAFTA countries.

Figure 4 shows how the estimated coefficients $\hat{\beta}_0 + \hat{\beta}_k$ vary across size deciles for exporters using either NAFTA or WTO. The gap between these coefficients reflects the sourcing distortion induced by RoO. While firms using NAFTA should mechanically be less likely to source from non-NAFTA due to compliance requirements, results show this distortion grows

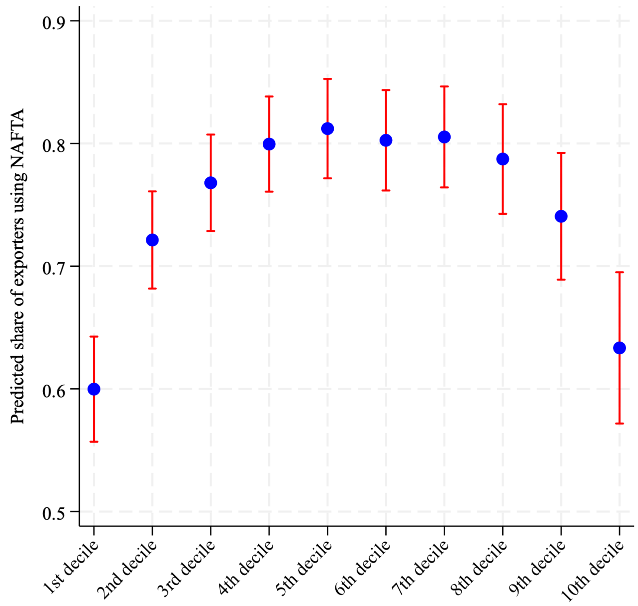


Figure 3: Predicted share of exporters using NAFTA by size decile.

with size, suggesting additional factors at play. Estimation results for Equation (2) appear in Table C.3 of Appendix C, along with the same robustness checks as for Equation (1).

We rationalize Empirical Facts 2 and 3 as stemming from Mexican exporters facing fixed costs of using NAFTA and sourcing from foreign countries. Fixed costs of using an FTA are well documented (e.g., Demidova et al. (2012), Cherkashin et al. (2015), Krishna et al. (2021)) and could be the result of labor, environmental, and health regulations, learning to use the agreement, and tracking input sourcing to present a *Certificate of Origin* at the border.[10] Similarly, sourcing from a foreign country entails finding a supplier, navigating domestic and foreign bureaucracy, and agreeing on product specifications, all of which we treat as fixed sourcing costs following Antras et al. (2017).

Firm size plays a key role in NAFTA use. Larger firms are better able to pay the fixed cost of using NAFTA, creating a positive link between size and NAFTA use. However, they are also more likely to profit from sourcing from foreign countries, raising the opportunity cost of complying with RoO and creating a negative link between size and NAFTA use.

These two fixed costs explain Figures 3 and 4. In the first, small firms cannot afford the fixed cost of using NAFTA, while large firms able to source globally find RoO restrictions too costly. In the second, small firms source little from abroad, so RoO have limited impact on their sourcing. As firms grow, their ability to engage in global sourcing increases, amplifying

[10] A document required under NAFTA to certify that a good being exported qualifies as an originating good, and thus qualifies for preferential tariff treatment.

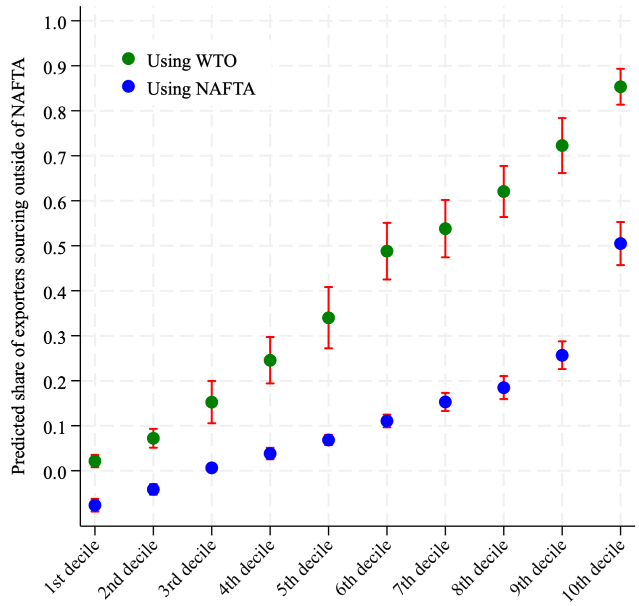


Figure 4: Predicted share of exporters sourcing outside of NAFTA by size decile.

the distortion RoO create in terms of sourcing from non-NAFTA countries.[11] This is our key mechanism: the opportunity cost of RoO rises with firm size, which our model captures through fixed costs of NAFTA use and foreign sourcing.

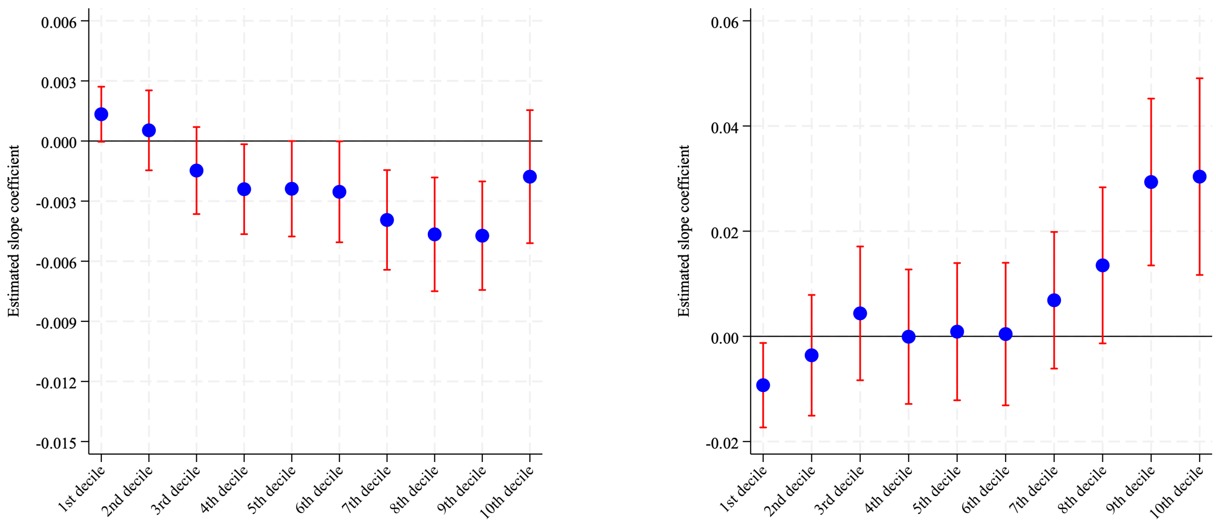
To provide evidence on these fixed costs, we exploit cross-product variation in RoO strictness and MFN tariffs across firm sizes. Without fixed costs, both the benefit and cost of using NAFTA would scale with size: the benefit being the ad-valorem tariff savings, and the cost the higher marginal cost from RoO compliance. This would imply homogeneous effects of RoO and MFN tariffs on NAFTA use across the size distribution. To test this, we estimate Equation (3), which allows the marginal effect of RoO and MFN tariffs on the use of NAFTA to vary across firm sizes:

$$\mathbb{N}_{ikjt} = \beta_1 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \alpha_1 \text{RoO}_j + \sum_{k=2}^{10} \alpha_k \mathbb{I}_{ikt} \times \text{RoO}_j + \gamma_1 \text{MFN}_j + \sum_{k=2}^{10} \gamma_k \mathbb{I}_{ikt} \times \text{MFN}_j + \iota_t + \epsilon_{ikjt} \quad (3)$$

where $\mathbb{N}_{ikjt} = 1$ if firm i of size decile k exporting product j at time t is using NAFTA to export, $\mathbb{I}_{ikt} = 1$ if the firm belongs to size decile k, and RoO_j and MFN_j are product j's RoO strictness and MFN tariff, respectively.

Figure 5 shows that the absolute value of slope coefficients $\hat{\alpha}_0 + \hat{\alpha}_k$ and $\hat{\gamma}_0 + \hat{\gamma}_k$ weakly

[11] The gap narrows for the 10th decile, possibly because very large import volumes make it profitable to pay fixed sourcing costs even under RoO.



(a) RoO strictness

(b) MFN tariffs

Figure 5: Marginal effects on the probability of using NAFTA to export.

increases with size, consistent with exporters facing fixed costs of using NAFTA and sourcing from foreign countries. Estimation results for Equation (3) are in Table C.5 of Appendix C.

For smaller firms, RoO and MFN tariffs have less impact on the probability of using NAFTA, as some are unable to pay fixed costs. Even with high MFN tariffs, a small firm may be unable to use NAFTA, and high RoO will not reduce its usage if it cannot source from foreign countries. As firms grow, fixed costs become less binding, amplifying the effects: higher MFN tariffs increase NAFTA use, while stricter RoO reduce it.

4 A Model of Rules of Origin and NAFTA Usage

We extend the partial equilibrium model in Antras et al. (2017) to incorporate RoO and the choice of using NAFTA. The model features CES preferences and monopolistically competitive final good producers (Melitz 2003), each sourcing a continuum of intermediates $\nu \in [0, 1]$ from competitive global suppliers (Eaton and Kortum 2002). Final good producers sell domestically and select their *sourcing strategy* —the set of countries from which they import intermediates —by paying fixed costs of sourcing. Our contribution is two-fold: (i) Mexican firms export final goods to the US, and (ii) firms choose whether to export under NAFTA or WTO. Using NAFTA requires compliance with RoO, which restrict sourcing by mandating that a share of inputs originate from NAFTA countries. To leverage product-level data on RoO strictness and MFN tariffs, the model incorporates multiple sectors and industries.

4.1 US Consumers

Let N be the set of NAFTA countries. We assume that consumers in the US value consumption of domestic good D and Mexican varieties ω across distinct industries and sectors.

$$U = D^{1-\eta} \prod_{s=1}^{S} \Big[\sum_{i=1}^{I_s} \Big(\int_{\omega \in \Omega_{si}} q_{si}(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \Big)^{\frac{\sigma(\epsilon-1)}{\epsilon(\sigma-1)}} \Big]^{\frac{\alpha_s \eta \epsilon}{(\epsilon-1)}} \tag{4}$$

Domestic good D represents all non-Mexican products. There are S sectors, each with I_s industries. Ω_{si} denotes the set of varieties from sector s, industry i, exported to the US. The elasticity of substitution across varieties within an industry is $\sigma > 1$, and across industries within a sector is $\epsilon > 1$, with $\sigma > \epsilon$ so demand is more elastic within industries. US consumers spend share $\eta \in [0, 1]$ of income on Mexican imports and $1-\eta$ on D, allocating share α_s of Mexican import expenditure to sector s.

$$\sum_s \alpha_s = 1$$

Consumers maximize their utility in Equation (4) subject to their budget constraint:

$$D + \sum_{s=1}^{S} \sum_{i=1}^{I_s} \Big(\int_{\omega \in \Omega_{si}} q_{si}(\omega) p_{si}(\omega) d\omega \Big) = E \tag{5}$$

where E denotes total US expenditure, and $p_{si}(\omega)$ is the price of variety ω in industry i and sector s. The domestic good D is the numeraire, so $p_{si}(\omega)$ is expressed relative to its price.

Solving the consumer's optimization problem, the demand in the US for variety ω is:

$$q_{si}(\omega) = \frac{\alpha_s \eta E P_{si}^{\sigma-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}} p_{si}(\omega)^{-\sigma} \tag{6}$$

where $P_{si} \equiv \Big(\int_0^1 p_{si}(\omega)^{1-\sigma} d\omega \Big)^{1/(1-\sigma)}$ is the Dixit-Stiglitz ideal price index for industry i in sector s. Demand for variety ω rises with US expenditure E, the share of spending on Mexican imports η, and the share α_s allocated to sector s, and it increases with the ideal price index while decreasing with price since $\sigma > 1$.

4.2 Mexican Exporters

To introduce RoO and the use of NAFTA, let $\kappa \in \{0, 1\}$ indicate whether a firm exports using NAFTA ($\kappa = 1$) or WTO ($\kappa = 0$). Let $\lambda(\omega) \in [0, 1]$ denote the share of inputs a firm must source from NAFTA countries if using NAFTA, and $\tau(\omega) \geq 0$ the ad-valorem tariff it would pay under WTO. Both are exogenous and variety-specific. Unlike Section 2, where RoO strictness is defined as the share of a product's input value restricted to NAFTA suppliers, here λ directly limits the share of a product's inputs used in production; Appendix E details when these definitions coincide. Using NAFTA entails a fixed cost $\zeta_{si} \in \mathbb{R}^{++}$, while sourcing from a foreign country j requires a fixed cost $f^j_{si}(\omega) \in \mathbb{R}^{++}$. For simplicity, we omit the ω index henceforth.

For clarity, we model firm behavior in four stages. First, a firm in industry i of sector s observes its λ, τ, and fixed costs ζ and f^j $\forall j$. Second, it decides whether to enter the US export market, learning its export-specific productivity ϕ only after paying a fixed entry cost υ. Third, it chooses to export using either NAFTA or WTO. Finally, it selects its sourcing strategy —the set of countries from which it will source inputs —based on its previous choice.

Final good producers use a continuum of inputs of measure one, specific to each industry i and sector s. Marginal costs vary across countries, industries, and sectors —for example, China may be more efficient in supplying inputs for industry i than for i'. The unit labor cost of producing an input, a^j_{si}, follows a Fréchet distribution:

$$F^j_{si}(a) = \exp(-T^j_{si} a^{-\theta})$$

where T^j_{si} is the aggregate productivity of country j in supplying inputs for industry i of sector s. Let $d^j_{si} > 1$ denote iceberg trade costs between Mexico and j, and w^j_{si} the wage per unit of labor in j for industry i of sector s. Intermediate input suppliers, domestic and foreign, are competitive and price at marginal cost. Thus, the price a Mexican firm pays for input $\nu \in [0, 1]$ is:

$$z_{si}(\nu, \kappa, \lambda, J) = \begin{cases} \min\limits_{j \in N \cap J} \{d^j_{si} a^j_{si}(\nu) w^j_{si}\} & \text{if } \nu \in [0, \kappa\lambda) \\ \\ \min\limits_{j \in J} \{d^j_{si} a^j_{si}(\nu) w^j_{si}\} & \text{if } \nu \in [\kappa\lambda, 1] \end{cases} \tag{7}$$

where $J(\phi, \kappa, \lambda, \tau)$ is the firm's sourcing strategy, i.e., the set of countries from which it can source inputs after paying fixed costs. Equation (7) states that a firm pays the lowest marginal cost among its sourcing countries for each input ν. RoO enter here: if exporting under NAFTA, inputs $\nu \in [0, \lambda)$ must be sourced exclusively from NAFTA countries in $N \cap J$, while inputs $\nu \in [\lambda, 1]$ can be sourced from any country in J.

The increase in firms' marginal cost when using NAFTA is proportional to the increase in expenditure in intermediate inputs induced by complying with RoO:

$$\int_0^{\kappa\lambda} \Big[\min_{j\in N\cap J}\{d^j_{si}a^j_{si}(\nu)w^j_{si}\} - \min_{j\in J}\{d^j_{si}a^j_{si}(\nu)w^j_{si}\}\Big] d\nu \tag{8}$$

Conditional on using NAFTA, Equation (8) shows RoO raise marginal costs, except when (i) NAFTA countries are the firm's most efficient suppliers, or (ii) the firm cannot pay fixed costs to source from non-NAFTA countries, i.e., $J \subseteq N$.

Within an industry, firms differ in export-specific productivity ϕ, drawn from a Pareto distribution with shape parameter χ. Given input prices in Equation (7), a firm with productivity ϕ faces marginal costs:

$$c_{si}(\phi,\kappa,\lambda,\tau,J) = \frac{1+(1-\kappa)\tau}{\phi}\left(\int_0^{\kappa\lambda} z_{si}(\nu)^{1-\rho}d^*\nu + \int_{\kappa\lambda}^{1} z_{si}(\nu)^{1-\rho}d\nu\right)^{1/(1-\rho)} \tag{9}$$

Equation (9) gives the marginal cost of producing and exporting. The benefit of using NAFTA arises when, if exporting under WTO ($\kappa = 0$), the firm must pay the MFN tariff $\tau \geq 0$. Marginal cost decreases with ϕ and is otherwise a CES aggregation over intermediate input prices, with $\rho > 1$ being the substitution parameter across them. The measure of inputs splits into two integrals: those restricted by RoO, $\nu \in [0,\lambda)$, and those freely sourced, $\nu \in [\lambda, 1]$, as their price distributions differ.

Using the Fréchet distribution for unit labor costs, we obtain that the marginal cost of a firm with productivity ϕ in industry i of sector s is:

$$c_{si}(\phi,\kappa,\lambda,\tau,J) = \frac{1}{\phi}\gamma^{-1/\theta}[1+(1-\kappa)\tau][\kappa\lambda\Psi_{si}(\phi)^{(\rho-1)/\theta} + (1-\kappa\lambda)\Phi_{si}(\phi)^{(\rho-1)/\theta}]^{\frac{1}{1-\rho}} \tag{10}$$

with:

$$\Psi_{si}(\phi) = \sum_{h\in N\cap J} T^h_{si}(d^h_{si}w^h_{si})^{-\theta} \quad \text{and} \quad \Phi_{si}(\phi) = \sum_{h\in J} T^h_{si}(d^h_{si}w^h_{si})^{-\theta}$$

We define $\Psi_{si}(\phi)$ as a firm's *NAFTA sourcing capability* and $\Phi_{si}(\phi)$ as its *total sourcing capability*. The term $T^h_{si}(d^h_{si}w^h_{si})^{-\theta}$ measures how attractive it is for a firm in industry i of sector s to source from country h, which we call the country's *sourcing potential* —the incentive to pay the fixed cost of sourcing from h. Finally, γ is the Gamma function evaluated at $(\theta + 1 - \rho)/\theta$.

The model predicts the share of inputs a firm sources from each country in its sourcing strategy J, with RoO introducing a wedge between NAFTA and non-NAFTA countries.

Using the Fréchet distribution properties, the share of inputs coming from a non-NAFTA country $j \in J \setminus N$ is:

$$x_{si}^{j}(\phi,\kappa,\lambda,J) = (1-\kappa\lambda)\frac{T_{si}^{j}(d_{si}^{j}w_{si}^{j})^{-\theta}}{\sum\limits_{h\in J} T_{si}^{h}(d_{si}^{h}w_{si}^{h})^{-\theta}} \tag{11}$$

while the share of inputs sourced from NAFTA country $j \in J \cap N$ is given by:

$$x_{si}^{n}(\phi,\kappa,\lambda,J) = \kappa\lambda\frac{T_{si}^{j}(d_{si}^{j}w_{si}^{j})^{-\theta}}{\sum\limits_{h\in N\cap J} T_{si}^{h}(d_{si}^{h}w_{si}^{h})^{-\theta}} + (1-\kappa\lambda)\frac{T_{si}^{j}(d_{si}^{j}w_{si}^{j})^{-\theta}}{\sum\limits_{h\in J} T_{si}^{h}(d_{si}^{h}w_{si}^{h})^{-\theta}} \tag{12}$$

In the model, with $\kappa = 1$, RoO raise the share of inputs sourced from NAFTA countries, regardless of their sourcing potential. If a firm exports under WTO ($\kappa = 0$) or faces no RoO ($\lambda = 0$), input shares reduce to the standard formulation in Eaton and Kortum (2002).

With monopolistic competition, firms set prices as a constant markup over marginal production and export costs, given US demand:

$$p_{si}(\phi,\kappa,\lambda,\tau,J) = \frac{\sigma}{\sigma-1}c_{si}(\phi,\kappa,\lambda,\tau,J) \tag{13}$$

Combining Equations (6), (10), and (13), a firm's operating profits for a choice of κ and J are given by:

$$\pi_{si}(\phi,\kappa,\lambda,\tau) = \phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa)\tau]^{1-\sigma}[\kappa\lambda\Psi_{si}(\phi)^{(\rho-1)/\theta} + (1-\kappa\lambda)\Phi_{si}(\phi)^{(\rho-1)/\theta}]^{\frac{1-\sigma}{1-\rho}} \tag{14}$$

where B_{si} represents US market demand for varieties from industry i of sector s:

$$B_{si} = \frac{1}{\sigma}\left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma}\left[\frac{\alpha_s\eta E P_{si}^{\sigma-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}}\right] \tag{15}$$

which the firm takes as given. Equation (14) shows that using NAFTA avoids paying MFN tariff τ but raises marginal costs, as λ shifts weight from total sourcing capability $\Phi_{si}(\phi)$ to NAFTA sourcing capability, and $\Psi_{si}(\phi) \leq \Phi_{si}(\phi)$. Without NAFTA or RoO, the expression reduces to the operating profits in Antras et al. (2017).

As described, firm choice occurs in stages: first, deciding whether to use NAFTA or WTO; then selecting the set of countries in its sourcing strategy. Assuming a firm has already chosen either NAFTA or WTO, it will then choose the sourcing strategy that maximizes its operating profits, subject to paying a fixed cost for each country from which they source:

$$\Pi_{si}(\phi,\kappa,\lambda,\tau) = \max_{I_{si}^j \in \{0,1\}_{j=1}^J} \pi_{si}(\phi,\kappa,\lambda,\tau,I^1,....,I^J) - w\sum_{j=1}^{J} I_{si}^j f_{si}^j(\phi) \tag{16}$$

where $I_{si}^j = 1$ if the firm includes country j in its sourcing strategy. Fixed costs of sourcing are firm-specific to capture heterogeneity; for example, firms near the US border likely face lower costs of sourcing from the US due to easier access to suppliers.

The optimization in Equation (16) is a combinatorial discrete choice problem, as firms select the country combination that maximizes profits given their prior choice of NAFTA or WTO. A firm using NAFTA has weaker incentives to source from non-NAFTA countries, since RoO require a share of inputs to come exclusively from NAFTA suppliers.

By backward induction, a firm chooses κ —NAFTA ($\kappa = 1$) or WTO ($\kappa = 0$) —to maximize profits subject to paying a fixed cost if it uses NAFTA:

$$\kappa^* = \arg\max_{\kappa\in\{0,1\}} \{\Pi_{si}(\phi,\kappa,\lambda,\tau) - \kappa w\zeta_{si}\} \tag{17}$$

where ζ_{si} is the fixed cost of using NAFTA, expressed in Mexican labor units, as are sourcing fixed costs. In order to close our model, free-entry into the US export market implies that expected export profits equal the fixed cost of market entry:

$$\int_{\tilde{\phi}_{is}}^{\infty} \left[\Pi_{si}(\phi,\kappa,\lambda,\tau) - \kappa(\phi)w\zeta_{si}\right] dG(\phi) = w\upsilon \tag{18}$$

where $G(\cdot)$ is the Pareto cdf, and $\tilde{\phi}_{si}$ is the productivity cutoff for the least productive firm in industry i of sector s that enters.

To summarize, firms observe their productivity, fixed costs, RoO if using NAFTA, and MFN tariffs if using WTO. They then decide whether to enter the US export market, choose NAFTA or WTO, and select a sourcing strategy. Finally, they meet demand, pricing at a constant markup over marginal cost.

4.3 Equilibrium

We assume the measure of non-exporting firms is large enough such that firms exporting to the US treat wages as exogenous; consistent with our model being one of partial equilibrium.

An equilibrium is a set of prices of varieties $p_{si}(\omega)$ and Mexican wages w such that:

1. Consumers maximize utility according to Equation (4) by choosing $q_{si}(\omega)$

2. Firms maximize profits in Equation (17) by choosing κ^* and J^* given $\{\lambda,\tau,f,\zeta\}$

3. Firms meet demand for their variety, given by Equation (6)

4. Expected profits of exporting to the US are zero, as in Equation (18)

This last condition results in the equilibrium number of firms of industry i from sector s actively exporting to the US being given by:

$$N_{si} = \frac{\alpha_s \eta E[1 - G(\tilde{\phi}_{si})]}{\sigma w \Big[\upsilon + \int_{\tilde{\phi}_{si}}^{\infty} \Big(\kappa(\phi)\zeta_{si} + \sum_{j \in J(\phi)} f_{si}^j(\phi) \Big) dG(\phi) \Big]} \times \frac{P_{si}^{1-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}} \tag{19}$$

Equation (19) shows that the number of Mexican firms exporting to the US in a given industry rises with the share of US expenditure on Mexican imports η, the sector's share α_s, and total US expenditure E. It falls with the elasticity of substitution σ, wage w (as fixed costs are in labor units), fixed costs of using NAFTA ζ and sourcing f, and the industry-level ideal price index $P_{si}^{1-\epsilon}$ since $\epsilon > 1$.

4.4 Input Purchases from Foreign Countries

Our model predicts Mexico's purchases of intermediate inputs from foreign countries. For a firm with productivity ϕ, purchases from country j can be expressed as a share of its operating profits:[12]

$$M_{si}^j(\phi) = (\sigma - 1) x_{si}^j(\phi, \kappa^*, \lambda, J^*) \pi_{si}^o(\phi, \kappa^*, \lambda, \tau) \tag{20}$$

where $x_{si}^j(\phi, \kappa, \lambda, J)$ is the share of inputs purchased from country $j \in J$ given optimal κ and J. As in Section 4.2, these shares differ for NAFTA and non-NAFTA countries due to the distortionary effects of RoO. For a non-NAFTA country $j \in J^* \setminus N$, input purchases are:

$$\begin{aligned} M_{si}^j(\phi) = {} & (\sigma - 1)\phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}(1 + (1 - \kappa^*)\tau)^{-\sigma} B_{si}(1 - \kappa^*\lambda)\Phi_{si}(\kappa^*)^{-1} T_{si}^j (d^j w_{si}^j)^{-\theta} \\ & \times \Big(\kappa^*\lambda\Psi_{si}(\kappa^*)^{(\rho-1)/\theta} + (1 - \kappa^*\lambda)\Phi_{si}(\kappa^*)^{(\rho-1)/\theta} \Big)^{\frac{1-\sigma}{1-\rho}} \end{aligned} \tag{21}$$

while input purchases from NAFTA country $j \in J^* \cap N$ follow:

[12] See Appendix G for the derivation.

$$M^j_{si}(\phi) = (\sigma-1)\phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}(1+(1-\kappa)\tau)^{-\sigma}B_{si}\left[\kappa^*\lambda\Psi_{si}(\kappa^*)^{-1}+(1-\kappa^*\lambda)\Phi_{si}(\kappa^*)^{-1}\right]T^j_{si}(d^jw^j_{si})^{-\theta}$$
$$\times\left(\kappa^*\lambda\Psi^*_i(\kappa)^{(\rho-1)/\theta}+(1-\kappa^*\lambda)\Phi^*_i(\kappa)^{(\rho-1)/\theta}\right)^{\frac{1-\sigma}{1-\rho}} \tag{22}$$

and $M^j_{si}(\phi)=0\ \forall j\notin J^*$. The model's predictions for a sector's input purchases from j are:

$$M^j_s = \sum_{i=1}^{I_s}\left[N_{si}\int_{\tilde{\phi}_{si}}^{\infty} M^j_{si}(\phi)dG(\phi)\right] \tag{23}$$

We cannot analytically characterize Equation (23) due to its high non-linearity. A firm's input purchases from a country depend on its sourcing potential and that of other countries in the firm's sourcing strategy, which is endogenous to its productivity and the RoO and MFN tariff it faces —factors that determine whether it exports under NAFTA or WTO. In practice, we simulate sectoral input purchases by first computing them at the firm level using Equation (20), then aggregating across simulated firms within each sector.

Equations (22) and (23) highlight the potentially adverse effects of a higher λ. Stricter RoO may not raise input purchases from NAFTA countries if firms instead opt for WTO. Even when firms comply and source a larger share from NAFTA, the resulting increase in marginal costs can outweigh the share gain, reducing overall imports of intermediates from a given NAFTA country.

4.5 Impact on the Use of NAFTA

This section examines how, in our model, NAFTA use depends on RoO, MFN tariffs, sourcing potentials, fixed costs, and cross-country comparative advantage. Without fixed costs of using NAFTA or sourcing from abroad, a firm would export under NAFTA if:

$$(1+\tau)\Phi^{-\frac{1}{\theta}} > \left[\lambda\Psi^{\frac{\rho-1}{\theta}}+(1-\lambda)\Phi^{\frac{\rho-1}{\theta}}\right]^{\frac{1}{1-\rho}}$$

which states that a firm will use NAFTA if the benefit —avoiding the MFN tariff —exceeds the cost, i.e., the marginal cost increase from RoO. Without fixed costs, Ψ and Φ are constant since firms include all countries in their sourcing strategy. It is convenient to rewrite this as:

$$LHS \equiv \frac{1-(1+\tau)^{1-\rho}}{\lambda} > 1-\left(\frac{\Psi}{\Phi}\right)^{\frac{\rho-1}{\theta}} \equiv RHS \tag{24}$$

Proposition 1. *The use of NAFTA is decreasing in RoO.*

The derivative of the RHS w.r.t. λ is 0 while that of the LHS is negative:

$$-\frac{[1-(1+\tau)^{1-\rho}]}{\lambda^2} < 0$$

making Inequality (24) less likely to hold. Higher RoO raise the price of inputs restricted to NAFTA sourcing, except in the unlikely case where non-NAFTA countries have zero sourcing potential for an industry.

Proposition 2. *The use of NAFTA is increasing in MFN tariffs.*

The derivative of the RHS w.r.t. τ is 0, while that of the LHS is positive:

$$\frac{\rho-1}{\lambda}(1+\tau)^{-\rho} > 0$$

making Inequality (24) more likely to hold. Since avoiding MFN tariffs is the main benefit of using NAFTA, higher tariffs increase the incentive to use it.

Proposition 3. *The use of NAFTA is increasing in the sourcing potential of a NAFTA country.*

The sourcing potential of a country is $T^j_{si}(d^j w^j_{si})^{-\theta}$, therefore, for NAFTA country $j \in N \cap J$:

$$\frac{\partial \Psi}{\partial T^j_{si}(d^j w^j_{si})^{-\theta}} = \frac{\partial \Phi}{\partial T^j_{si}(d^j w^j_{si})^{-\theta}} = 1$$

and thus the derivative of the LHS is 0, while that of the RHS is negative given $\Phi > \Psi$:

$$\frac{1-\rho}{\theta}\left(\frac{\Psi}{\Phi}\right)^{\frac{\rho-1}{\theta}-1}\left[\frac{\Phi-\Psi}{\Phi^2}\right] < 0$$

making Inequality (24) more likely to hold. A higher sourcing potential of NAFTA countries relative to non-NAFTA ones lowers the opportunity cost of RoO. For instance, if Mexico is already the best source for a firm's inputs, RoO do not raise its marginal cost.

Proposition 4. *The use of NAFTA is decreasing in the sourcing potential of non-NAFTA countries.*

For non-NAFTA country $j \in J \setminus N$:

$$\frac{\partial \Psi}{\partial T^j_{si}(d^j w^j_{si})^{-\theta}} = 0$$

and thus the derivative of the LHS is 0, while that of the RHS is positive:

$$\frac{1-\rho}{\theta}\Big(\frac{\Psi}{\Phi}\Big)^{\frac{\rho-1}{\theta}-1}\Big[-\frac{\Psi}{\Phi^2}\Big]>0$$

making Inequality (24) less likely to hold. Higher sourcing potential of non-NAFTA countries raises the opportunity cost of RoO. For example, if China is the best source for a firm's inputs, forcing it to buy from NAFTA countries is costly.

Propositions 3 and 4 are only neccessarily true in the absence of fixed costs, which complicate the analysis. For example, if Mexico's sourcing potential rises, Proposition 3 predicts greater NAFTA use. However, the resulting lower marginal cost may raise firm revenue enough to cover the fixed cost of sourcing from a non-NAFTA country, reducing the incentive to use NAFTA.

Proposition 5. *The use of NAFTA is decreasing in the fixed cost of using NAFTA to export, and increasing in the fixed cost of sourcing from a foreign country.*

This follows because if a firm cannot source inputs from a foreign country, RoO will not alter its sourcing strategy. For example, if it cannot import from China, a rule prohibiting such sourcing is irrelevant.

Proposition 6. *The stronger comparative advantage is across countries, the larger the effects RoO have on the use of NAFTA.*

Parameter θ is the shape parameter of the Fréchet distribution for a country's unit labor costs. Lower θ implies greater cost variability and thus, stronger comparative advantage across countries. From Equation (14), lower θ magnifies differences between firms' NAFTA and total sourcing capabilities, raising the opportunity cost of complying with RoO. Intuitively, θ acts as an elasticity of substitution across sourcing countries: lower values make countries more complementary, increasing the cost of sourcing restrictions.

5 Taking the Model to the Data

This section outlines our three-step estimation procedure. First, we take several parameters from the literature to reduce computational burden. Second, we estimate countries' sourcing potentials at the industry level —i.e., the attractiveness of sourcing inputs from a given country —by regressing firm-level input shares on industry-country fixed effects. Third, we use the Simulated Method of Moments together with a Simulated Annealing algorithm to estimate fixed costs and US market demand, matching simulated moments to their data counterparts following Eaton et al. (2011).

We map industries in the model to HS 2-digit codes, aggregated into six sectors: Agriculture and Foods, Minerals and Chemicals, Skins and Textiles, Mining, Manufacturing, and Others.[13] Within sectors, industries differ in their RoO strictness, MFN tariffs, number of firms, and sourcing potentials. All are taken directly from the data, except sourcing potentials, estimated in 5.2.

For computational tractability, we group input-sourcing regions into: Mexico, US and Canada, China, Europe,[14] and Rest of the World. This keeps the SMM estimation feasible, as the set of possible sourcing strategies grows exponentially with the number of countries. Unlike Antras et al. (2017), profits in our model are not necessarily supermodular in productivity due to the non-linearities stemming from NAFTA choice and RoO. Thus, we cannot reduce the problem's dimensionality as in Jia (2008) and must compute firm profits for each possible sourcing strategy (see Appendix H).

5.1 Parametrization

We take several parameters from the literature, listed in Table 1: the elasticity of substitution across final goods σ, across industries ϵ, the shape parameter θ of the Fréchet distribution for unit labor costs, and the shape parameter χ of the Pareto distribution for firm productivity. We set $\rho = 1.05$, the substitution parameter across intermediate inputs, implying an elasticity of substitution $1/\rho$, so inputs are complementary in production.

Parameter	Value	Source
σ	3.85	Antras et al. (2017)
ϵ	3.00	Broda and Weinstein (2006)
θ	1.79	Antras et al. (2017)
χ	4.25	Melitz and Redding (2015)

Table 1: Parameters from the literature.

[13] Agriculture and Foods: HS Sections I-IV; Minerals and Chemicals: HS Sections V-VII; Skins and Textiles: HS Sections VIII-XII; Mining: HS Sections XIII-XV; Manufacturing: HS Sections XVI-XVIII; Others: HS Sections XIX-XXII.

[14] This group includes the ten largest exporters of intermediates to Mexico from the continent: DEU, GBR, FRA, ITA, RUS, ESP, NLD, TUR, SWE, and POL.

5.2 Estimation of Sourcing Potentials

We estimate countries' sourcing attractiveness following Antras et al. (2017). As shown in Section 4.2, the share of inputs a firm sources from region j when using WTO or facing no RoO is:

$$x^j_{si}(\phi) = \frac{T^j_{si}(d^j_{si}w^j_{si})^{-\theta}}{\sum\limits_{h\in J} T^h_{si}(d^h_{si}w^h_{si})^{-\theta}}$$

Normalizing Mexico's sourcing potential to one, i.e. $T^{MEX}_{si}(d^{MEX}_{si}w^{MEX}_{si})^{-\theta} = 1$, and taking the ratio between the sourcing potential of region j and that of Mexico results in:

$$\frac{x^j_{si}(\phi)}{x^{MEX}_{si}(\phi)} = T^j_{si}(d^j_{si}w^j_{si})^{-\theta} \tag{25}$$

Taking logs from both sides and assuming an idiosyncratic measurement error ϵ^j_{si} yields:

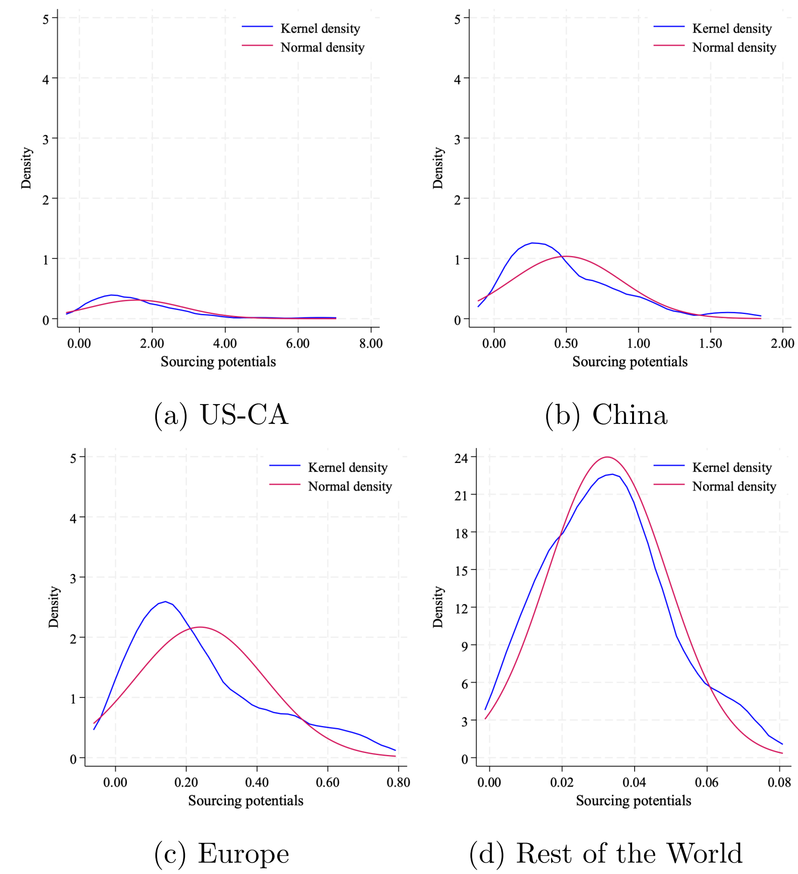
$$lnx^j_{si}(\phi) - lnx^{MEX}_{si}(\phi) = lnT^j_{si}(d^j_{si}w^j_{si})^{-\theta} + \epsilon^j_{si}(\phi)$$

We estimate this with OLS, regressing firm-level input shares on industry-region fixed effects. The sample is restricted to firms using WTO or facing no RoO, since for them the ratio of input shares depends only on region j's sourcing potential, as in Equation (25).

We infer input shares since domestic sourcing by Mexican firms is unobserved. Using the DRC Tables for each product's input composition, we assume: (i) unimported inputs are sourced from Mexico, and (ii) imported inputs are sourced entirely from the observed foreign region. For example, if a product is 60% input A and 40% input B, and input B is imported from China, the shares are 60% Mexico and 40% China, with all others at 0%. Provided measurement error is idiosyncratic across firms and industries, this yields unbiased estimates of each foreign region's sourcing attractiveness.

The above estimation measures how well a region supplies inputs for a given exporting industry (e.g., China supplying electrical components). To capture the overall benefit of sourcing from a region, we redefine sourcing potentials at the importing industry level (e.g., attractiveness of China to Mexico's automotive industry). Using each final product's input composition, we compute a weighted average of input-specific sourcing potentials. For example, if China's potential is 0.4 for *Input A* and 0.6 for *Input B*, and a Mexican industry uses each input equally, China's industry-specific potential is 0.5.

Figure 6 reports industry-specific sourcing potentials by region. Averaging across industries, sourcing potentials are 1.55 for US-CA, 0.49 for China, 0.24 for Europe, and 0.03 for the Rest of the World. Thus, a firm sourcing from all countries has a total sourcing capability 231% higher than one sourcing only from Mexico.[15] In contrast, Antras et al. (2017) estimate only a 19% gain, likely because the US is relatively better than Mexico at supplying inputs to its domestic firms, making our estimates larger.



(a) US-CA (b) China

(c) Europe (d) Rest of the World

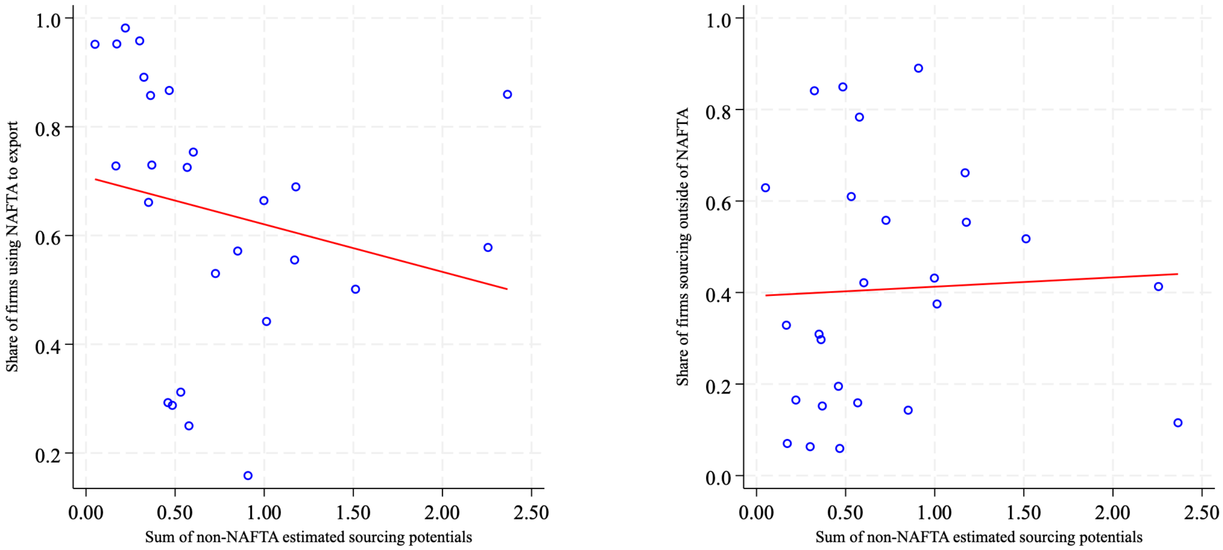
Figure 6: Kernel Densities for Estimated Sourcing Potentials.

The lowest potential sourcing capability is in *Preparations of meat, or fish, or crustaceans* (HS Chapter 16), only 13% higher than sourcing solely from Mexico. The highest is *Plastics and articles thereof* (HS Chapter 39), with an 826% gain. Full industry-region results are in Appendix I.

The sourcing potential measure captures how attractive regions are as input sources, so forgoing high-potential non-NAFTA regions raises marginal costs. Thus, in industries where these regions are more attractive: (i) NAFTA usage should be lower, due to the higher opportunity cost of RoO, and (ii) the share of firms sourcing from them should be higher, as paying fixed costs is profitable. Results align with this: Panel 7a shows a negative correlation between sourcing potentials and NAFTA use, while Panel 7b shows a positive correlation

[15] By construction, a firm sourcing solely from Mexico has a sourcing capability of 1.

with sourcing from non-NAFTA countries.



(a) NAFTA Usage

(b) Non-NAFTA Sourcing

Figure 7: Correlations with estimated non-NAFTA sourcing potentials.

5.3 Estimation of Fixed Costs and US Market Demand

We structurally estimate sector-specific fixed costs of using NAFTA, fixed costs of sourcing from each foreign region, and US market demand. For tractability, we assume these parameters are sector-specific rather than industry-sector specific. As in Section 4.2, the fixed cost of using NAFTA is constant across firms within a sector, and fixed sourcing costs from foreign countries follow:

$$f_s^j(\phi) \sim \text{Log-normal}(\mu_s^j, \delta_s^j) \tag{26}$$

The location parameter $\mu_s^j \in (-\infty, +\infty)$ is region-sector specific to capture differences across countries (e.g., proximity, language) and sectors (e.g., ease of finding suppliers abroad, input customization). We set $\delta_s^j = \sqrt{\log 3}$, as μ_s^j already influences both the mean and variance of fixed costs. Since firms in our data, as described in Section 5.2, always source from Mexico, its fixed sourcing cost is set to zero.

For every sector s, we separately estimate the fixed cost of using NAFTA, the location parameter of fixed costs of sourcing, and US market demand:

$$\xi_s = [\zeta_s, \mu_s^{US-CA}, \mu_s^{CHN}, \mu_s^{EUR}, \mu_s^{ROW}, B_s]$$

Estimation follows Eaton et al. (2011), using Simulated Method of Moments with a Simulated Annealing algorithm, which combines random exploration of the parameter space with directed searches for parameters that minimize the objective function.[16] Let x_s denote sectoral data and ξ_s the sector-specific parameters. We estimate ξ_s by minimizing:

$$\min_{\xi_s} ||e(\tilde{x_s}, x_s|\xi_s)||$$

The moment error function $e(\tilde{x_s}, x_s|\xi_s)$ is expressed as the percentage points difference between the vectors of simulated and data moments:

$$e(\tilde{x_s}, x_s|\xi_s) = \hat{m}(\tilde{x_s}|\xi_s) - m(x_s)$$

where $m(\cdot)$ is a set of R moments, and $\tilde{x}_s$ is simulated data from the model under parameters ξ_s. Using the L^2 norm, SMM minimizes the sum of squared errors:

$$\hat{\xi}_s = \arg\max_{\xi_s} e(\tilde{x_s}, x_s|\xi_s)^T I_R e(\tilde{x_s}, x_s|\xi_s)$$

with I_R being an $R \times R$ identity matrix. Informed by the empirical facts described in Section 3, the set of sector-specific moments we include to identify the true parameter vector ξ_s are:

1. Share of firms using NAFTA to export: This moment identifies the fixed cost ζ_s, as changes in it directly affect the number of firms able to pay for NAFTA usage. We also include the share of firms in the lowest quintile of RoO strictness, for whom this fixed cost should be the main factor in the export decision.

2. Share of firms sourcing from each region: Given sourcing potentials, higher sourcing costs reduce the share of firms importing from a region. This extensive margin helps identify the location parameter for the distribution of fixed sourcing costs by region.

3. Median firm-level exports: We use sector-level median exports to identify US market demand. In our model, productivity and demand are isomorphic in revenue, and with a common productivity distribution across sectors, US demand maps one-to-one to firm-level exports.

Our simulation of the model for a given ξ_s proceeds as follows: (i) For each industry i in sector s, draw N_{si} productivities from a Pareto distribution with shape parameter χ, where N_{si} (firms exporting to the US) comes from the data. (ii) For each sector s, draw $N_s =$

[16]The algorithm's *temperature* controls the acceptance rate of points that do not reduce the objective function, adapting the Metropolis-Hastings algorithm in Metropolis et al. (1953).

Sector	ζ_s/X	f_s^{CHN}/X	f_s^{EUR}/X	f_s^{US-CA}/X	f_s^{ROW}/X	B_s
Agriculture and Foods	1.10	25.45	26.50	26.14	6.16	0.05
Minerals and Chemicals	0.18	14.82	5.37	30.71	0.60	0.01
Skins and Textiles	0.51	8.21	4.79	23.07	0.41	0.02
Mining	0.25	22.49	37.47	41.78	2.76	0.01
Manufactures	1.60	2.88	2.43	12.09	0.76	0.03
Others	1.17	8.98	4.58	20.37	0.64	0.02
Average	0.80	13.81	13.52	25.69	1.89	0.02

Table 2: Estimates for Fixed Costs and US Market Demand.

$\sum_{i=1}^{I_s} N_{si}$ fixed costs of sourcing from each region using Equation (26), where I_s is the number of industries in s. Industries differ in RoO, MFN tariffs, and sourcing potentials; firms within an industry differ in productivity. (iii) By backward induction and fixing NAFTA or WTO, determine for each firm the set of sourcing countries maximizing profits net of fixed sourcing costs, per Equation (16). (iv) Compare profits under NAFTA and WTO (Equation (17)) and assign each firm to the more profitable option. Using firms' optimal NAFTA choice and sourcing strategy, compute predicted input shares, prices, exports, profits, and intermediate imports via Equations (11) - (20).

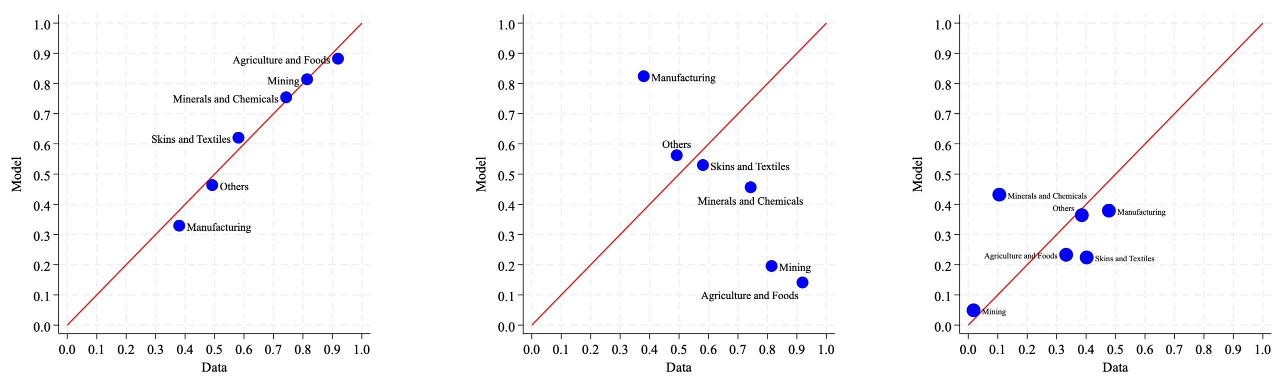
We present the estimation results in Table 2. For the fixed costs of using NAFTA ζ and sourcing f, we report transformations of the SMM point estimates for easier interpretation —specifically, average fixed costs as a share of total export revenue X. For example, in *Agriculture and Foods*, sourcing from China costs on average 25.45% of revenue. For US market demand B, we report the direct point estimates. Compared to Antras et al. (2017), where US firms face average demand $B = 0.12$, our results imply Mexican exporters face 8.3% - 41.7% of US firms' market demand. Full estimation results are in Appendix J.

Our results show substantial heterogeneity across sectors in the fixed cost of using NAFTA, likely reflecting differences in regulation and learning to use the FTA. Exporting under NAFTA requires presenting a *Certificate of Origin* at the border, detailing the origin of all intermediate inputs. Sectors with more complex products or supply chains, such as *Manufacturing*, face higher costs to track input sourcing. Regulatory compliance also varies: for example, capital-intensive sectors like *Mining* may face lower costs to meet NAFTA's labor requirements. Second, fixed costs of sourcing vary across both regions and sectors. Cross-region differences likely reflect factors such as language, cultural proximity, and the strength of Mexico's business ties with foreign partners. For a given region, sectoral differences may stem from how easily Mexican exporters can find suppliers or from the degree of input customization required. For instance, foreign suppliers for *Manufacturing* and *Skins and Textiles* may be more globally connected. These differences matter because RoO effects are heterogeneous: in sectors where few firms can afford NAFTA's fixed cost, RoO have little impact, whereas in sectors with low foreign sourcing costs, RoO can be very costly by restricting access to efficient global suppliers. Lastly, US market demand is highest for *Agriculture and Foods* and *Manufactures*, consistent with these sectors accounting for most of Mexico's export value to the US in our sample.

5.4 Fit of the Model

This section evaluates the model's fit at the sectoral and aggregate levels. Figure 8 compares model predictions with data for: (a) the share of firms using NAFTA to export, (b) the share sourcing inputs outside NAFTA, and (c) median firm-level exports. The x-axis shows the data moments and the y-axis the simulated ones, so points closer to the 45-degree line indicate a better fit for each sector.

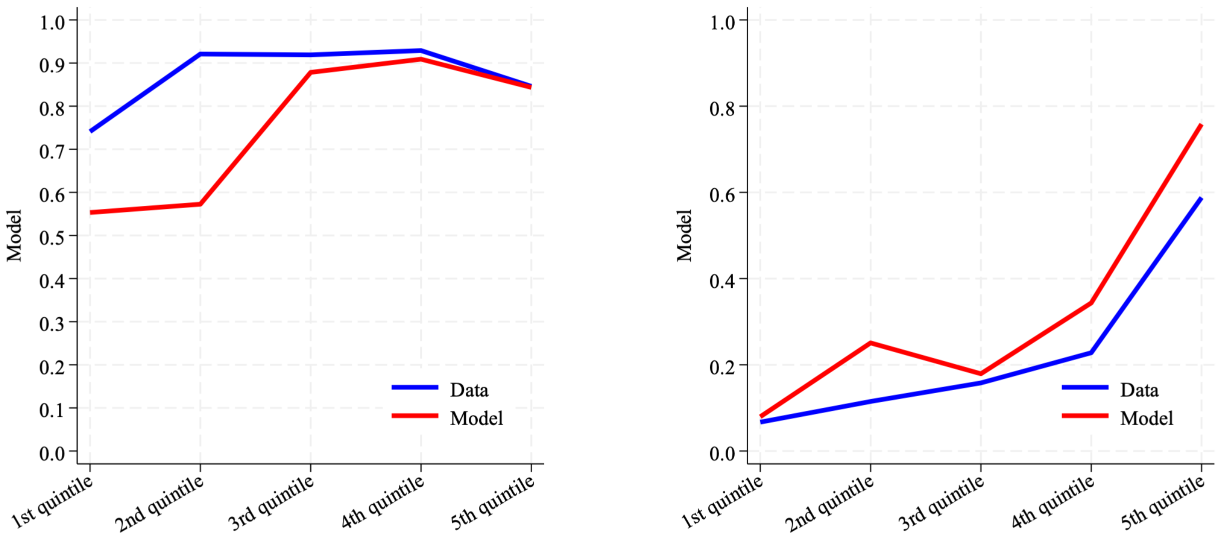
Across sectors, the model matches the sectoral use of NAFTA and median firm-level exports, except for the latter in *Minerals and Chemicals*. For the share of firms sourcing from outside NAFTA, results are mixed: the model underpredicts for *Agriculture and Foods* and *Mining*, and overpredicts for *Manufacturing*. Overprediction could stem from overestimating non-NAFTA sourcing potential in these sectors, which would also explain discrepancies in regional input shares shown in Figure K.2. This is less likely due to underestimated fixed sourcing costs, as entry rates match well in Figure K.1 in Appendix K.



(a) NAFTA Usage (b) Non-NAFTA Sourcing (c) Median Exports

Figure 8: Sectoral fit of the model.

At the aggregate level, depicted in Figure 9, we group firms into size quintiles. Panel 9a shows the model reproduces the inverse U-shaped relationship between NAFTA use and firm size. As in Empirical Fact 2, the model predicts small firms use NAFTA less due to high



(a) NAFTA Usage

(b) Non-NAFTA Sourcing

Figure 9: Aggregate fit of the model.

fixed costs, while large firms face higher opportunity costs of RoO compliance from greater foreign sourcing.

Regarding sourcing outside NAFTA, Panel 9b shows the model replicates its positive relationship with firm size. Larger firms are more likely to pay the fixed cost of sourcing from foreign countries, while NAFTA use reduces incentives to source from non-NAFTA regions. As in the data, the model predicts that although the largest firms use NAFTA less than medium-sized firms, a greater share source from non-NAFTA countries, reflecting their ability to profitably pay these costs despite RoO restrictions.

6 Effects of Trade Policies

This section presents the counterfactual scenarios. Using our product-level data and given that our model captures key trade policy dimensions —most notably RoO and tariffs —, we simulate policy-relevant scenarios. The first counterfactual examines a 20 percentage-point increase in RoO strictness, a policy instrument originally intended to prevent trade diversion by avoiding preferential access through minimal processing by non-members, but now often aimed at maximizing intra-regional trade. The second considers a 20 percentage-point increase in MFN tariffs. For each, we assess effects on trade flows between US-CA and Mexico and the US price index for Mexican imports, paying particular attention to variation across the firm-size distribution, the central mechanism determining the opportunity cost of

RoO compliance. To isolate this mechanism, we also simulate a no-dispersion case, removing firm-level heterogeneity within industries.

We then explore two additional exercises. Section 6.2 examines how results change when firms face adjustment frictions, limiting their ability to reconfigure sourcing strategies after policy shocks. These frictions may reflect short-run constraints or persistent differences in flexibility across firms and sectors. Section 6.3 analyzes how the industry-level share of inputs sourced from US-CA, and the value of US-CA intermediate exports to Mexico, vary with RoO strictness, producing industry-specific Laffer curves for these outcomes.

These exercises abstract from other margins of adjustment not captured in our counterfactuals. For instance, we treat countries' sourcing potentials as exogenous, though they may respond to changes in RoO or tariffs. As Ornelas and Turner (2023) note, RoO can spur investment in domestic or regional industries, especially in high-productivity sectors, which may help explain why policymakers adopt stricter rules to protect local producers from foreign competition. Moreover, given the model's limitations, our counterfactuals do not account for firms possibly shifting assembly to avoid stricter RoO, as pointed out in Cadot et al. (2006) and Head et al. (2024), nor for general equilibrium effects such as US firms' responses or adjustments in Mexican prices and wages.

6.1 Changes in Trade Policy

We simulate a uniform 20 p.p. increase in RoO strictness across industries. Aggregating across sectors, Table 3 shows that NAFTA use falls by 5.68%, sourcing from non-NAFTA regions by 15.99%, and both average exports and firm profits by 4.72% and 4.10%, respectively. The US-CA price index for Mexican imports rises by 1.61%, reflecting higher marginal costs as firms either remain under NAFTA but switch to less efficient regional suppliers or switch to WTO and pay tariffs. While regional content in Mexican exports rises slightly (0.63%), Table 5 shows US-CA intermediate exports to Mexico fall by 3.89%, indicating stricter RoO need not increase regional purchases once NAFTA exit and scale effects are considered.

These aggregate effects mask the firm-size heterogeneity central to our paper. Table 4 shows stricter RoO reduce NAFTA use across all quintiles, but nonlinearly: the second and fifth quintiles see the largest drops (8.93% and 12.07%), while the first and middle quintiles are more resilient. The mechanism is twofold. Small firms are already constrained by NAFTA's fixed cost, leaving little scope for intensive-margin adjustment. Large firms, with broader sourcing strategies, face the highest cost, as stricter RoO force substitution from efficient non-NAFTA suppliers, making NAFTA less attractive and pushing some to switch to WTO status. The probability of sourcing from non-NAFTA regions also falls across

Counterfactual	+20 pp RoO	+20 pp RoO, same productivity	+20 pp Tariff	+20 pp Tariff, same productivity
NAFTA Usage	-5.68	-5.82	30.11	17.54
Share Non-NAFTA Sourcing	-15.99	-19.63	-8.23	-6.96
Average Firm Exports	-4.72	-3.86	-1.50	-1.49
Average Firm Profits	-4.10	-2.28	-2.13	-2.15
US Price Index	1.61	2.66	-0.50	-0.08
Share NAFTA Value	-13.16	-10.18	23.47	17.25
Average Regional Content	0.63	0.68	2.12	2.26
Tariff Revenue	35.91	38.40	-98.89	-97.78

Table 3: Percent changes in aggregate moments.

all quintiles, with especially large declines in the fourth (26.73%) and fifth (15.64%).

Yet, despite the mechanical push toward NAFTA suppliers, Table 5 shows US-CA imports would fall by 3.89%. This reflects two forces: a scale effect from higher marginal costs and the exit of some firms from NAFTA, removing RoO-driven demand for US-CA inputs. Thus, we observe a slight rise in average regional content but a decline in US-CA intermediate export values. The same table highlights the extensive-margin effect: entry into non-NAFTA regions falls sharply —13.68% for China, 15.11% for Europe, and 19.14% for the Rest of the World —while NAFTA entry changes little.

We next examine a 20 percentage-point increase in MFN tariffs. Table 3 shows NAFTA usage rising sharply by 30.11%, with regional input shares up 2.12%. This shift comes at the expense of non-NAFTA sourcing, which falls 8.23%, including steep drops from China (26.23%) and Europe (23.65%). The US import price index for Mexican goods declines slightly (0.50%), but Mexican firms face higher intermediate costs from reconfigured sourcing. As a result, average profits fall 2.13% and total exports dip modestly. In short, the tariff hike deepens NAFTA integration and lowers import prices, but at the cost of firm profitability and export volume.

Size heterogeneity again drives the adjustment. Table 4 shows NAFTA usage rising most among the bottom two quintiles (67.78% and 68.80%), where firms were previously marginal users due to fixed costs; the tariff hike raises the benefit of avoiding WTO status enough to overcome these barriers. At the top, switching is smaller (18.52% in the fifth quintile) as large firms are more exposed to RoO-related efficiency losses. Non-NAFTA sourcing contracts most in the middle of the size distribution (22.29% in the third quintile), with smaller declines at the top, consistent with large firms retaining diversified sourcing. Given

Counterfactual	+20 pp RoO	+20 pp RoO, same productivity	+20 pp Tariff	+20 pp Tariff, same productivity
NAFTA Usage 1st	-1.79	-2.96	67.78	30.27
NAFTA Usage 2nd	-8.93	-11.52	68.80	40.27
NAFTA Usage 3rd	-3.27	0.00	13.29	0.00
NAFTA Usage 4th	-2.38	-2.71	9.82	7.15
NAFTA Usage 5th	-12.07	-14.15	18.52	19.63
Non-NAFTA Sourcing 1st	-13.08	-7.31	-21.84	-43.69
Non-NAFTA Sourcing 2nd	-4.26	-7.79	-18.92	-9.83
Non-NAFTA Sourcing 3rd	-14.61	0.00	-22.29	0.00
Non-NAFTA Sourcing 4th	-26.73	-13.80	-3.97	-2.10
Non-NAFTA Sourcing 5th	-15.64	-31.48	-1.89	-2.19

Table 4: Percent change in NAFTA Usage and Non-NAFTA Sourcing by size quintile.

Counterfactual	+20 pp RoO	+20 pp RoO, same productivity	+20 pp Tariff	+20 pp Tariff, same productivity
Entry Rate CHN	-13.68	-15.60	-13.21	-14.16
Entry Rate EUR	-15.11	-17.52	-13.67	-15.94
Entry Rate US-CA	-1.18	-1.39	1.18	1.11
Entry Rate ROW	-19.14	-23.23	-9.88	-10.26
Input Share MXN	0.62	0.76	1.74	1.99
Input Share CHN	-7.99	-7.07	-36.17	-35.91
Input Share EUR	-13.33	-16.78	-26.12	-25.36
Input Share US-CA	0.66	0.34	3.73	3.29
Input Share ROW	-20.98	-20.41	-12.54	-9.53
Domestic Purchases	-2.06	-1.47	1.48	0.88
Imports from CHN	-15.23	-15.88	-26.23	-31.15
Imports from EUR	-17.08	-20.42	-23.65	-28.46
Imports from US-CA	-3.89	-3.95	3.18	2.38
Imports from ROW	-24.66	-22.52	-9.25	-7.25

Table 5: Percent changes in region-level moments.

stricter RoO, size-sector interactions should add another layer: in sectors where NAFTA suppliers are efficient, large firms should face lower RoO compliance costs and maintain NAFTA usage, while in sectors reliant on non-NAFTA inputs, the same policy shock should prompt sharp NAFTA exit and larger declines in US-CA imports.

Removing productivity variation makes firms homogeneous within an industry, though size differences persist due to RoO, MFN tariffs, fixed costs, US demand, and sourcing potentials. With a 20 p.p. RoO increase, NAFTA usage and non-NAFTA sourcing vary less across sizes, but larger firms still reduce NAFTA use more. For all size groups, non-NAFTA sourcing declines, US-CA imports fall, the US import price index rises, and profits drop. With a 20 p.p. MFN tariff increase, removing productivity dispersion compresses size differences in adjustment. In the baseline, larger firms could better absorb tariffs and keep non-NAFTA sourcing, while smaller firms shifted into NAFTA; with identical productivity, NAFTA usage rises more uniformly. The US price index falls slightly and profits decline moderately. Overall, productivity heterogeneity amplifies size-dependent responses —without it, the qualitative effects of policy shocks persist, but firm-level impacts converge.

6.2 Adjustment Frictions in Sourcing Strategies

The counterfactuals in Section 6.1 assume firms can instantly re-optimize their sourcing strategies and export regime after a policy shock. In reality, supplier switching is constrained by contracts, certification, relationship-specific investments, and organizational bottlenecks. To capture these rigidities while keeping the model tractable, we introduce a stochastic adjustment mechanism: firms can adjust their suppliers with Bernoulli probability p.

After the shock, each firm receives a random draw from a Bernoulli distribution with probability p. A draw of one allows full reoptimization: adding or dropping sourcing locations and, given the joint optimality of export regime and sourcing strategy, switching between NAFTA and WTO. A draw of zero freezes the sourcing strategy but still allows changing the export regime. Thus, p is the share of firms able to reconfigure immediately, and $1 - p$ the share locked into their pre-shock sourcing strategies.

This formulation has three interpretations. First, varying p from 0 to 1 could span the adjustment path from short-run (no firm adjusts) to long-run (all reconfigure). Second, across firm sizes, low p could reflect large exporters with high switching costs from deep supplier integration, while high p could approximate smaller, more agile firms. Third, across sectors, standardized-input industries could effectively be high-p, while those reliant on customized inputs or specialized certifications could be low-p. Figures 10 and 11 show how NAFTA usage, non-NAFTA sourcing, Mexican exports to the US, and imports of intermediates from

US-CA vary with p under the two trade policy shocks.

Consider Figure 10, where RoO strictness rises by twenty percentage-points. With p near zero, most firms cannot reconfigure, so compliance is met mechanically with existing NAFTA suppliers. NAFTA usage jumps relative to the frictionless case, even though many firms would prefer to exit NAFTA ex post, while non-NAFTA sourcing stays flat; as p rises, this share falls as firms either drop NAFTA or shed non-NAFTA suppliers to meet the stricter requirement. US exports from Mexico fall most sharply when p is low, as locked-in sourcing drives up marginal costs; higher p lets more firms reconfigure toward cheaper mixes, softening but not reversing the export loss. The effect on US-CA intermediate exports hinges on p: near zero, they rise as firms meet higher content rules with regional inputs; as p grows, NAFTA exits and higher marginal costs erode US-CA sales.

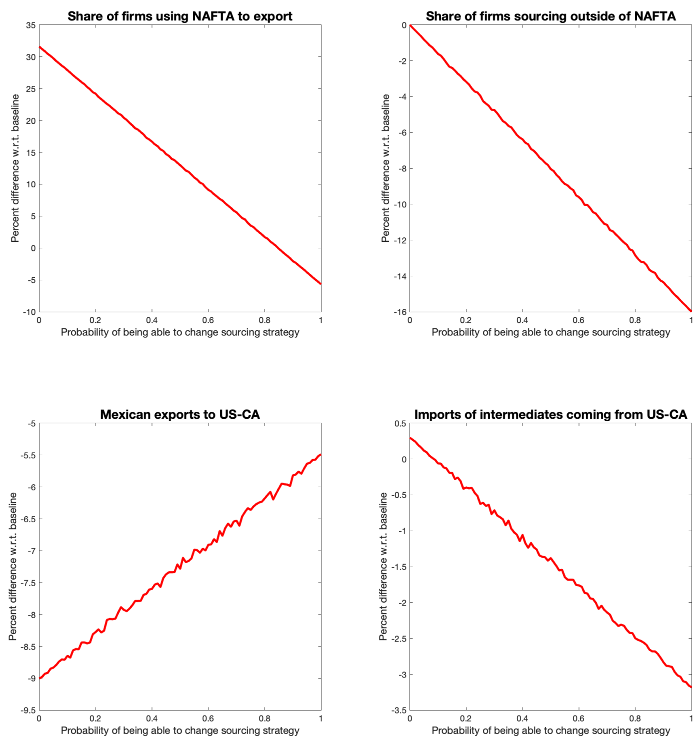


Figure 10: Percent changes in moments given a 20 pp increase in RoO Strictness.

Figure 11 shows the effects of a 20 percentage-point MFN tariff hike. NAFTA usage surges when p is near zero, as most firms cannot reconfigure and the only way to avoid tariffs is to use NAFTA given their pre-shock sourcing. Many firms near the RoO threshold find compliance cheaper than paying the tariff, driving the jump. As p rises, some firms can reoptimize —even if that means paying the tariff —so the NAFTA increase is smaller. Mexican exports and US-CA intermediate exports change little with p, as tariff savings lower marginal costs but replacing efficient non-NAFTA suppliers with costlier NAFTA ones raises them, leaving trade volumes roughly unchanged. Thus p shapes sourcing composition and NAFTA prevalence, but not aggregate trade with US-CA under the tariff shock.

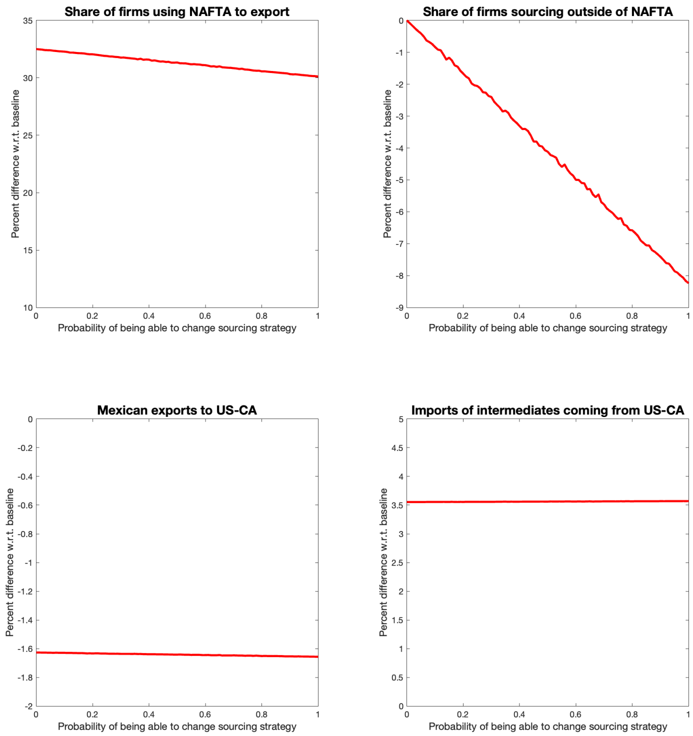


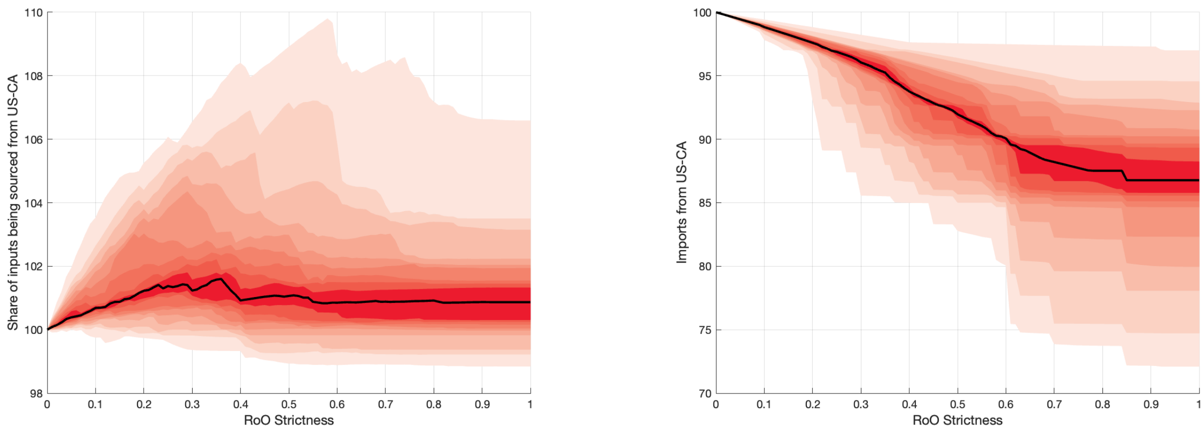
Figure 11: Percent changes in moments given a 20 pp increase in MFN Tariffs.

Taken together, this exercise offers a tractable way to study adjustment without a full dynamic model. It nests short and long-run responses, maps naturally to firm-size and sectoral rigidity, and provides comparative statics for key outcome variables of policy interest. Crucially, it suggests that regional content and bilateral trade values can diverge sharply depending on the economy's position along the adjustment path.

6.3 Industry-Level Laffer Curves

Building on the mechanisms discussed in Section 6.1, we now turn to the industry level to examine how RoO strictness shapes regional integration. Following the spirit of Head et al. (2024), who examine how regional content shares vary with the content requirements, we map for each industry the relationship between RoO strictness and two outcomes of policy interest: the share of US-CA inputs in Mexican final goods and US-CA intermediate exports to Mexico.

Figure 12 plots these outcomes by industry as functions of RoO strictness. In the left panel, the share of US-CA intermediates in Mexican production follows a hump-shaped



(a) Share of US-CA inputs

(b) US-CA exports of intermediates

Figure 12: Laffer Curves as a function of RoO strictness.

pattern: it rises with stricter RoO, peaks, and then falls. This aligns with a Laffer-curve logic, in which there exists a positive RoO strictness that maximizes regional content. By contrast, the right panel shows that the value of US-CA intermediate exports to Mexico declines monotonically with RoO strictness.

The hump-shaped share curves in the left panel reflect substitution at low to moderate RoO strictness and exit at higher levels. Initially, tighter RoO push NAFTA-using firms to substitute toward US-CA suppliers to meet content requirements, raising the share of regional inputs. Beyond a threshold, however, the efficiency loss from replacing non-NAFTA suppliers outweighs tariff savings, prompting some firms —especially those relying on efficient non-NAFTA inputs —to reduce output or switch to WTO, which lowers regional content. By contrast, the right panel's value measure falls monotonically: stricter RoO shrink both the number of NAFTA users and export scale. Even when regional content rises, higher marginal costs cut total import volumes from US-CA, so the efficiency loss dominates the compositional gain.

The divergence between share and value measures highlights that higher regional content does not imply greater trade in value terms. Focusing only on regional content risks overstating the benefits of stricter RoO. Sectoral heterogeneity matters: industries where NAFTA suppliers are close in efficiency to non-NAFTA ones should be able to sustain higher RoO before the share declines, while sectors reliant on efficient non-NAFTA inputs should peak at lower strictness. Yet the value measure falls in every case —stricter RoO never increase the total value of US-CA intermediate exports to Mexico.

The industry patterns in Figure 12 show that focusing only on regional input shares can mask trade-volume losses. Stricter RoO may raise content shares among participating firms, but often prompt exit or downsizing by those reliant on efficient non-NAFTA suppliers. This reduces Mexican exports and, in turn, the total value of US-CA intermediate exports. That volume falls in all industries —even where input shares peak —underscores the trade-off between maximizing content shares and sustaining trade scale. Policy design should weigh deeper regional linkages against retaining large, globally competitive firms whose exit can offset integration gains.

7 Conclusions

The core message of the paper is that firm size shapes how RoO affect NAFTA use. Mexican exporters are likely to face two types of fixed costs: (i) using NAFTA, which deters small firms, and (ii) sourcing from foreign suppliers, which results in larger firms having a relatively higher opportunity cost of RoO compliance. Stricter RoO thus have opposite effects across the size distribution: small firms rarely switch into NAFTA, while large firms are more likely to exit when compliance requires costly sourcing changes.

Restrictive trade policies —whether through stricter RoO or higher MFN tariffs —reshape sourcing patterns, prices, and trade flows, with aggregate effects masking substantial heterogeneity across firms and sectors. Two main forces drive these differences: (i) the exporter size distribution and (ii) the relative efficiency of NAFTA versus non-NAFTA suppliers. In sectors where NAFTA suppliers are relatively efficient, stricter RoO can raise regional content shares; where non-NAFTA suppliers are more efficient, they tend to lower NAFTA usage and reduce sourcing efficiency. Adjustment costs further shape these responses, influencing both the speed at which firms reconfigure their sourcing after a policy change and the variation across firms and sectors in their ability to do so. Industry-level results highlight a fundamental trade-off: while stricter RoO can increase the share of regional inputs, they consistently reduce the overall scale of regional trade because the higher compliance costs from replacing efficient non-NAFTA suppliers outweigh the gains from greater regional sourcing.

These results yield two key policy considerations. First, sourcing efficiency declines most in sectors where NAFTA suppliers face a comparative disadvantage, with losses likely larger in less diversified economies such as Mexico. Second, stricter RoO do not necessarily deepen regional integration in value terms: even when regional content shares increase, the total value of bilateral trade falls as less efficient sourcing and lower output offset the gains from higher regional content.

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Appendix

A Direct Requirement Coefficients in the HS

According to the BEA, Direct Requirement Coefficients "show the amount of inputs purchased directly to produce one dollar of output," i.e., the exact input composition of each final product, with coefficients summing to one across a product's inputs. As noted in Section 2, the DRC Tables are reported in the BEA's own product classification, whereas the rest of our data is in HS. While correspondence tables between these systems are readily available, the mapping is many-to-many: a BEA code can match multiple HS 6-digit codes, and vice versa.

Whenever a BEA input code corresponds to multiple HS codes, we allocate its direct requirement coefficient uniformly across all corresponding HS codes. This yields a correspondence in which outputs remain in BEA classification, inputs are in HS, and input shares sum to one for each output. When an HS output code matches multiple BEA codes, we take the average of their direct requirement coefficients. The resulting table provides the exact input composition of each HS 6-digit product.

To illustrate how we construct HS input composition tables, we use the following example. Suppose the DRC Tables look like Table A.1.

BEA Code Output	BEA Code Inputs	DR Coefficient
A	C	0.6
A	D	0.4
B	C	0.8
B	D	0.2

Table A.1: Example of Direct Requirement Coefficients Tables.

The BEA-HS correspondence follows the structure illustrated in Table A.2. The match is many-to-many: for example, BEA code A maps to HS codes 1 and 2, while HS code 1 maps to BEA codes A and B.

The first step is to construct a match where outputs are defined in BEA and inputs in HS. Using Table A.2, we expand Table A.1 by uniformly distributing each BEA input's direct requirement coefficients across its corresponding HS codes. The result, shown in Table A.3, ensures that for each BEA output the DR coefficients across HS inputs sum to one. We also retain BEA-HS input pairs with implied DR coefficients of zero —for instance, HS code 5 does not correspond to BEA code C.

BEA Code	HS Code
A	1
A	2
B	1
B	3
C	3
C	4
D	4
D	5

Table A.2: Example of the BEA-HS Correspondence.

Next, we sum the DR coefficients for each BEA output-HS input pair, and then match BEA outputs to HS outputs using the correspondence in Table A.2. For example, HS output code 1 corresponds to both BEA codes A and B. Table A.4 illustrates the resulting DR coefficient table for HS output code 1.

Finally, we average the DR coefficients for each HS output-input pair; this step is unnecessary when an HS code corresponds to a single BEA code. Table A.5 presents the resulting input composition of HS output code 1 in terms of its HS input codes. Ensuring that these HS-level DR coefficients sum to one is crucial, as our RoO strictness measure is computed as a weighted average of the input value restricted under RoO for each final product.

B RoO Strictness and MFN Tariffs

Tables B.1 and B.2 report average RoO strictness and MFN tariffs by HS section. RoO strictness is highest in *Textiles* and *Stone, Cement, Glass*, and lowest in *Arms and Ammunitions* and *Chemical Products*. Under NAFTA, high RoO strictness arises when a large share of a product's inputs must be sourced exclusively from within the region, or when the restricted inputs account for a disproportionately large share of production. In Table B.2, the highest MFN tariffs are found in *Textiles*, *Footwear and Headgear*, and *Stone, Cement, Glass*, while the lowest are in *Vehicles and Aircraft* and *Arms and Ammunitions*.

Tables B.3 and B.3 list the top 20 HS 4-digit headings by average RoO strictness and MFN tariffs, respectively. Most entries in both rankings belong to the *Textiles* industry. The overlap already hints at a strong positive correlation between RoO strictness and MFN tariffs, suggesting that Mexican firms face a trade-off when deciding whether to use NAFTA for their exports to the US. Figure B.1 plots the empirical distributions of RoO strictness

BEA Output	BEA Inputs	HS Inputs	Adjusted BEA DR Coefficient
A	C	3	0.3
A	C	4	0.3
A	C	5	0.0
A	D	3	0.0
A	D	4	0.2
A	D	5	0.2
B	C	3	0.4
B	C	4	0.4
B	C	5	0.0
B	D	3	0.0
B	D	4	0.1
B	D	5	0.1

Table A.3: BEA Outputs - HS Inputs Correspondence.

HS Output	HS Inputs	Adjusted HS DR Coefficient
1	3	0.3
1	4	0.5
1	5	0.2
1	3	0.4
1	4	0.5
1	5	0.1

Table A.4: HS Outputs - HS Inputs Correspondence.

and MFN tariffs at the HS 6-digit level for products in our sample, showing substantially greater dispersion in RoO strictness than in the corresponding MFN tariffs.

C Regression Tables and Robustness Checks

In this appendix, we present the estimation tables and robustness checks for the empirical facts documented in the paper. Table C.1 reports the results for the probability of using NAFTA to export, both from Equation 1 and from an alternative specification that replaces the product-level RoO strictness and MFN tariff controls with industry fixed effects. These fixed effects capture unobserved heterogeneity across industries, including variation in their RoO and MFN tariffs.

HS Output	HS Inputs	Adjusted HS DR Coefficient
1	3	0.35
1	4	0.50
1	5	0.15

Table A.5: Input Composition of HS Code 1.

HS Section	RoO Strictness
Animals Live/Products	20.60
Vegetable Products	21.26
Beverages and Tobacco	14.54
Chemical Products	0.14
Plastic and Rubber	2.44
Skins and Leather	8.72
Wood Products	0.96
Textiles	65.90
Footwear and Headgear	13.63
Stone, Cement, Glass	25.87
Precious Stones and Metals	15.89
Base Metals	1.28
Machinery and Electrical	0.36
Vehicles and Aircraft	10.25
Precision Instruments	0.93
Arms and Ammunitions	0.01
Misc. Manufactured Articles	3.87

Table B.1: Average RoO strictness by HS section.

Figure C.1 plots the predicted share of firms using NAFTA by firm-size decile, setting RoO strictness and MFN tariffs to zero for simplicity. The blue line replicates Figure 3; the red line adds industry fixed effects; and the green line controls for RoO strictness, MFN tariffs, and industry fixed effects jointly. The inverse U-shape relationship between NAFTA usage and firm size is robust to these controls, indicating it is not driven by industry-level heterogeneity. As an additional robustness check, we estimate the following specification:

$$\mathbb{N}_{ikjt} = \beta_0 + \beta_1 \text{Size}_{it} + \beta_2 \text{Size}_{it}^2 + \alpha_1 \text{RoO}_j + \alpha_2 \text{MFN}_j + \iota_t + \epsilon_{ikjt} \tag{27}$$

where $\mathbb{N}ikjt = 1$ if firm i of size k exporting product j at time t uses NAFTA, and Sizeit is a proxy for firm i's size at time t. We use two proxies: (i) size percentiles, as defined in Section 3, and (ii) the log of total exports. In alternative specifications, we either control for RoO strictness and MFN tariffs or include industry fixed effects. Table C.2 reports the results.

In our data, we cannot identify whether firms belong to Global Value Chains (GVCs) or operate as Maquilas. These firms may have limited discretion over whether to use NAFTA or WTO status, or over their sourcing strategy and compliance with RoO. Including them in

HS Section	MFN Tariff
Animals Live/Products	3.77
Vegetable Products	10.66
Beverages and Tobacco	7.87
Chemical Products	4.71
Plastic and Rubber	3.92
Skins and Leather	9.38
Wood Products	3.68
Textiles	10.62
Footwear and Headgear	10.96
Stone, Cement, Glass	11.90
Precious Stones and Metals	5.40
Base Metals	3.47
Machinery and Electrical	3.36
Vehicles and Aircraft	2.66
Precision Instruments	4.40
Arms and Ammunitions	2.70
Misc. Manufactured Articles	5.77

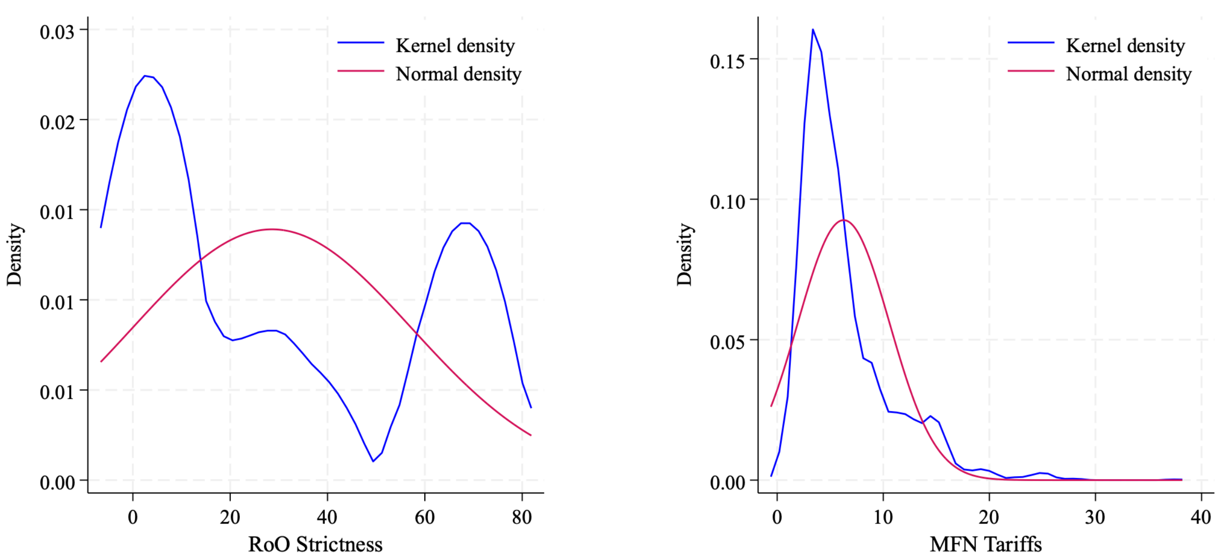
Table B.2: Average MFN tariff by HS section.

HS Heading	Heading Description	RoO Strictness
0305	Fish, dried, salted or in brine; smoked fish...	75.27
6304	Furnishing articles...	75.06
6302	Bed linen, table linen, toilet linen and kitchen linen	75.06
6303	Curtains (including drapes) and interior blinds...	75.06
5805	Tapestries; hand-woven...	74.31
6205	Shirts; men's or boys'...	70.15
6208	Singlets and other vests, slips...	70.15
6201	Overcoats, car-coats, capes...	70.15
6203	Suits, ensembles, jackets, blazers...	70.15
6215	Ties, bow ties and cravats...	70.15
6217	Clothing accessories...	70.15
6206	Blouses, shirts and shirt-blouses...	70.15
6211	Track suits, swimwear and other garments...	70.15
6207	Singlets and other vests, underpants...	70.15
6213	Handkerchiefs...	70.15
6214	Shawls, scarves, mufflers...	70.15
6210	Garments made up of fabrics...	70.15
6202	Coats; women's or girls' overcoats...	70.15
6204	Suits, ensembles, jackets, dresses...	70.15
6209	Garments and clothing accessories; babies'...	66.21

Table B.3: Top 20 HS headings by average RoO strictness.

HS Heading	Heading Description	MFN Tariff
6401	Footwear; waterproof, with outer soles...	37.50
8704	Vehicles; for the transport of goods...	24.12
6404	Footwear; with outer soles of rubber...	20.83
6111	Garments and clothing accessories...	20.82
0703	Onions, shallots, garlic, leeks...	20.00
6112	Track suits, ski suits and swimwear...	19.54
1507	Soya-bean oil and its fractions...	19.10
6209	Garments and clothing accessories; babies'...	17.46
0712	Vegetables, dried; whole, cut, sliced...	16.92
6105	Shirts; men's or boys'...	16.54
6106	Blouses, shirts and shirt-blouses; women's or girls'...	16.45
6203	Overcoats, car-coats, capes, cloaks...	16.23
6101	Coats; men's or boys' overcoats...	15.90
0709	Vegetables; fresh or chilled...	15.33
6109	T-shirts, singlets and other vests...	15.27
6212	Brassieres, girdles, corsets, braces, suspenders...	15.11
5112	Woven fabrics of combed wool...	15.10
5809	Fabrics, woven; of metal thread and metallised yarn...	14.90
5513	Woven fabrics of synthetic staple fibres...	14.67
5408	Woven fabrics of artificial filament yarn...	14.57

Table B.4: Top 20 HS headings by average MFN tariff.



(a) RoO Strictness

(b) MFN Tariffs

Figure B.1: Empirical distributions at the HS 6-digit level.

	(1) Pr(NAFTA)	(2) Pr(NAFTA)	(3) Pr(NAFTA)
d2	0.122***	0.102***	0.104***
	(7.26)	(3.97)	(3.90)
d3	0.168***	0.133***	0.134***
	(8.41)	(4.22)	(4.10)
d4	0.200***	0.137***	0.136***
	(9.68)	(4.01)	(3.85)
d5	0.212***	0.145***	0.144***
	(10.01)	(4.34)	(4.16)
d6	0.203***	0.137***	0.136***
	(9.37)	(4.00)	(3.86)
d7	0.206***	0.132***	0.135***
	(9.55)	(3.86)	(3.79)
d8	0.188***	0.121***	0.127***
	(8.25)	(3.40)	(3.40)
d9	0.141***	0.0467	0.0490
	(5.49)	(1.03)	(1.07)
d10	0.0336	0.0263	0.0504
	(1.07)	(0.53)	(0.99)
RoO Strictness	-0.00184***		0.00237*
	(-4.74)		(2.09)
MFN Tariff	0.00580**		-0.0114***
	(3.02)		(-5.49)
Constant	0.600***	0.610***	0.632***
	(27.45)	(20.98)	(12.99)
HS Chapter F.E.	✗	✓	✓
Observations	103,917	163,111	140,671

t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.1: Regression Output for the Probability of using NAFTA

	(1) Pr(NAFTA)	(2) Pr(NAFTA)	(3) Pr(NAFTA)	(4) Pr(NAFTA)
Size percentile	0.0102***	0.00659***		
	(11.37)	(7.52)		
Percentile sq	-0.0000982***	-0.0000584***		
	(-10.75)	(-7.65)		
RoO Strictness	-0.00186***		-0.00198***	
	(-4.78)		(-5.09)	
MFN Tariff	0.00557**		0.00563**	
	(2.91)		(2.94)	
Log of exports			0.153***	0.0881***
			(10.18)	(6.56)
Log of exports sq			-0.00663***	-0.00379***
			(-9.46)	(-6.01)
Constant	0.565***	0.673***	-0.0925	0.320***
	(24.30)	(30.14)	(-1.17)	(4.40)
HS Chapter F.E.	✗	✓	✗	✓
Observations	103,917	70,785	103,917	70,785

t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.2: Robustness Checks for the Probability of using NAFTA

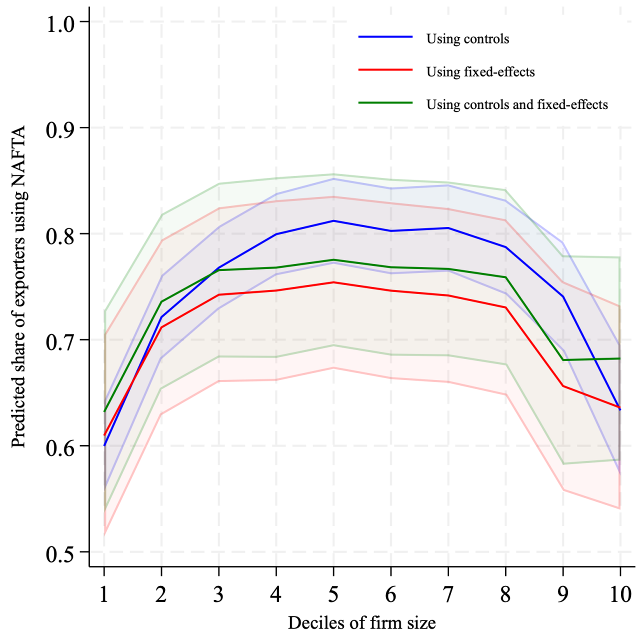


Figure C.1: Predicted share of exporters using NAFTA by size decile.

the sample could bias our estimates. For example, if a Maquila must export using NAFTA, its probability of doing so will be unaffected by RoO strictness, introducing a downward bias in the estimated marginal effect of RoO strictness on NAFTA usage. The same logic applies to GVC firms, which may lack sufficient market power to determine their export regime or sourcing strategy independently. To address this concern, we conduct two additional robustness checks: (i) re-estimate Equation 1 excluding the automotive and textiles industries,[17] as anecdotal evidence suggests these sectors hosted a large share of Mexican GVC and Maquila firms; and (ii) re-estimate the same equation, excluding one industry at a time, to ensure that our inverse U-shape relationship is not driven by any single sector.

Figure C.2 reports the inverse U-shape between NAFTA usage and firm size after excluding either the Textiles or Automotive industries. To assess whether our findings are driven by any single sector, Figure C.3 plots the *convex hull* of predicted NAFTA usage by size decile from a set of estimations in which we sequentially remove one industry at a time. For each estimation, we predict the share of firms using NAFTA in each decile, identify the lowest and highest predicted values, and shade the area between them.

These results confirm the robustness of the inverse U-shaped relationship between NAFTA usage and firm size. The pattern holds when controlling for product-level incentives to use

[17]For textiles, we drop all observations corresponding to HS Chapters 61, 62, and 63. For automobiles, we drop HS Chapter 87.

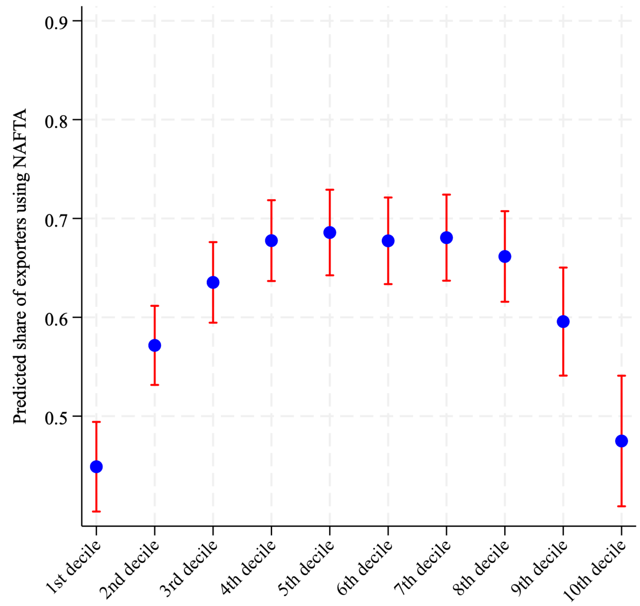


Figure C.2: Predicted share of exporters using NAFTA removing Textiles and Automotive.

NAFTA, adding industry fixed effects to capture industry-level heterogeneity, employing alternative firm-size proxies, and excluding industries —such as Textiles and Automotive —that may be disproportionately affected by GVC or Maquila participation.

In terms of how firm size affects the probability of sourcing inputs outside of NAFTA, the estimation results for Equation 2 are reported in Table C.3. We conduct the same robustness checks as for the probability of using NAFTA to export. First, Table C.4 presents estimates for the following equation:

$$\mathbb{S}_{ist} = \beta_0 + \beta_1 \text{Size}_{it} + \beta_2 \text{Size}_{it}^2 + \iota_{st} + \epsilon_{ikjt} \tag{28}$$

where $\mathbb{S}ist = 1$ if firm i in industry s sources inputs outside of NAFTA at time t, and Sizeit again denotes alternative proxies for firm size. Results are robust to using these different measures. We then re-estimate the relationship under two additional robustness checks: (i) excluding observations from the Textiles or Automotive industries, and (ii) removing one industry at a time from the sample.

Results for (i) are shown in Figure C.4, which confirms that our empirical finding on the distortion caused by RoO is not driven by the Textiles or Automotive industries. For (ii), we estimate Equation 29, but rather than estimating the relationship separately for firms using NAFTA and those using WTO, we explicitly control for the effect of using NAFTA to export and allow this effect to vary by firm size:

	(1) Firms using WTO	(2) Firms using NAFTA
2nd decile	0.0507***	0.0351***
	(4.89)	(6.64)
3rd decile	0.131***	0.0830***
	(5.46)	(11.07)
4th decile	0.224***	0.115***
	(8.43)	(11.50)
5th decile	0.318***	0.145***
	(9.02)	(13.94)
6th decile	0.467***	0.187***
	(14.03)	(15.74)
7th decile	0.517***	0.229***
	(15.12)	(15.38)
8th decile	0.599***	0.261***
	(19.42)	(15.09)
9th decile	0.701***	0.333***
	(21.55)	(16.80)
10th decile	0.832***	0.581***
	(38.05)	(21.94)
Constant	0.0215**	-0.0765***
	(3.06)	(-10.63)
HS Chapter F.E.	✓	✓
Observations	13,542	57,209

t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.3: Regression Output for the Probability of Non-NAFTA sourcing.

	(1) Firms using WTO	(2) Firms using NAFTA	(3) Firms using WTO	(4) Firms using NAFTA
Size percentile	0.0118***	-0.000312		
	(9.84)	(-0.27)		
Percentile squared	-0.0000239*	0.0000604***		
	(-1.99)	(4.55)		
Log of exports			0.110***	-0.0948***
			(6.41)	(-4.86)
Log of exports sq			-0.000688	0.00747***
			(-0.93)	(7.28)
Constant	-0.000222	-0.0159	-0.645***	0.237**
	(-0.01)	(-0.90)	(-7.11)	(2.65)
HS Chapter F.E.	✓	✓	✓	✓
Observations	47,056	112,964	47,056	112,964

t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.4: Robustness Checks for the Probability of Non-NAFTA Sourcing

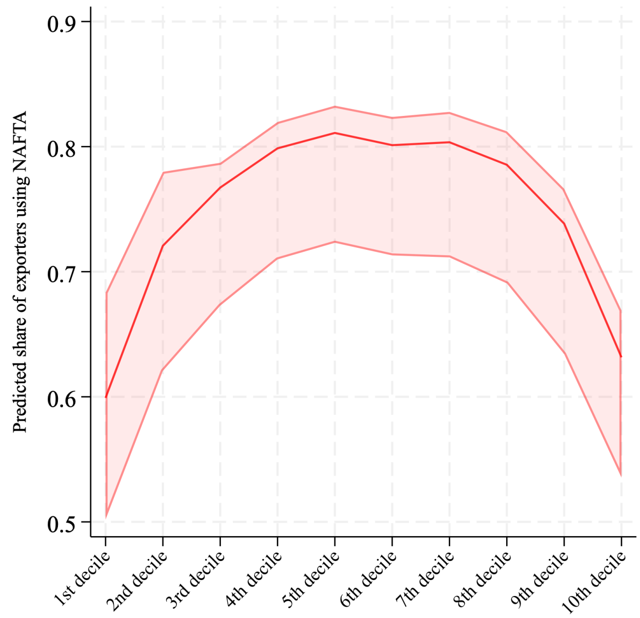


Figure C.3: Predicted share of exporters using NAFTA removing one industry at a time.

$$\mathbb{S}_{ikst} = \beta_0^n + \sum_{k=2}^{10} \beta_k^n \mathbb{I}_{ikt} + \sum_{k=1}^{10} \alpha_k^n \mathbb{I}_{ikt} \mathbb{N}_{ikt} + \iota_{st}^n + \epsilon_{ijt} \tag{29}$$

Figure C.5 plots the mean and convex hull of α_k across size deciles. Our sample contains 53 industries, yielding 53 sets of $\alpha_{k\,k=1}^{n\,10}$. For a given decile k, the central line represents the average across all estimations:

$$\bar{\alpha}_k = \frac{1}{53} \sum_{n=1}^{53} \alpha_k^n$$

The shaded area spans from the lowest estimated effect, $\alpha_k^{\min} = \min_n \alpha_k^n$, to the highest, $\alpha_k^{\max} = \max_n \alpha_k^n$, for each decile k. Thus, the shaded region does not measure the magnitude of the distortion itself, but rather the range of predicted distortions obtained when one industry is removed at a time. The distortion induced by RoO corresponds to the vertical distance between the zero line and any point within the convex hull of predictions.

These results confirm the robustness of our empirical finding of a U-shaped relationship between firm size and the distortion in non-NAFTA sourcing induced by RoO.

Table C.5 reports the full estimation results for Equation 3. We omit the dummy for the first decile, both for the intercept and for the interactions with RoO strictness and MFN tariffs, so all coefficients should be interpreted relative to the first decile.

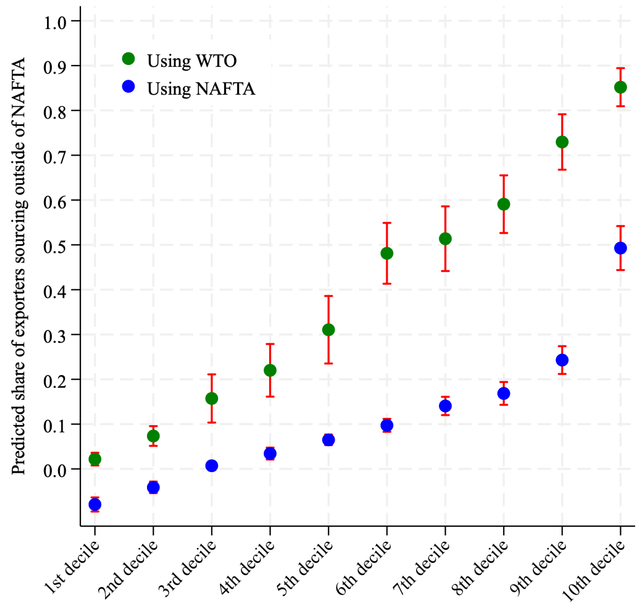


Figure C.4: Predicted non-NAFTA sourcing removing Textiles and Automotive.

D Additional Empirical Facts

This section presents additional empirical facts from the data. Figure D.1 plots the time series of the share of firms using NAFTA to export, by destination. Overall, the share remained stable over the sample period, though some seasonality is apparent. This seasonality likely reflects shifts in the composition of Mexican exports throughout the year; for example, vegetables —exported mostly in winter —are more intensively exported under NAFTA. The figure also shows that NAFTA usage rates are similar for exports to the US and to Canada, although volatility is higher for the latter.

Consistent with our structural model, there is substantial industry-level heterogeneity in both NAFTA usage and sourcing from non-NAFTA countries, as shown in Figures D.2 and D.3, respectively. Industries are defined at the HS 1-digit level of aggregation. For NAFTA usage, this heterogeneity should arise from differences in either the benefits or the costs of using NAFTA. On the benefit side, industries vary in the MFN tariffs they would face if exporting under WTO. On the cost side, they differ in both their RoO requirements and the fixed costs of using NAFTA. Our model incorporates these features by directly using industry-level RoO and MFN tariffs and allowing the fixed cost of using NAFTA to be industry-specific.

Turning to sourcing from outside NAFTA, heterogeneity again reflects differences in the relative benefits and costs of doing so. On the benefit side, industries vary in how attractive

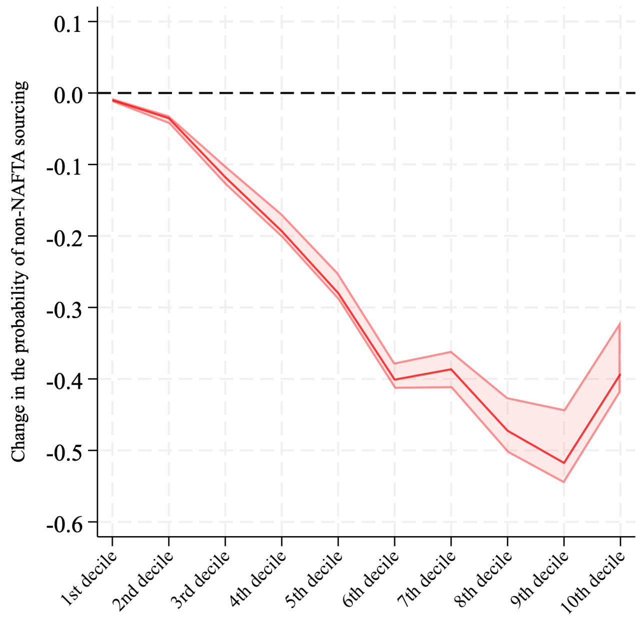


Figure C.5: Distortion in non-NAFTA sourcing removing one industry at a time.

non-NAFTA suppliers are, depending on global patterns of comparative advantage. For example, if Mexican firms in a given industry can source efficiently from the US, there is little incentive to import from non-NAFTA regions. On the cost side, industries differ in how easily they can access foreign suppliers, which depends on factors such as the nature of their inputs and the degree of their global integration. In our model, we capture this heterogeneity by allowing both the attractiveness of sourcing from foreign regions and the fixed costs of doing so to vary across industries.

	(1) Pr(NAFTA)
2nd decile	0.0913** (2.93)
3rd decile	0.122** (2.94)
4th decile	0.225*** (5.19)
5th decile	0.229*** (4.96)
6th decile	0.230*** (4.98)
7th decile	0.214*** (4.75)
8th decile	0.152** (2.98)
9th decile	-0.0534 (-0.77)
10th decile	-0.223** (-3.28)
RoO Strictness	0.00134 (1.91)
2nd decile × RoO Strictness	-0.000805 (-1.09)
3rd decile × RoO Strictness	-0.00281** (-3.27)
4th decile × RoO Strictness	-0.00374*** (-4.13)
5th decile × RoO Strictness	-0.00372*** (-3.75)
6th decile × RoO Strictness	-0.00387*** (-3.59)
7th decile × RoO Strictness	-0.00528*** (-4.98)
8th decile × RoO Strictness	-0.00600*** (-4.73)
9th decile × RoO Strictness	-0.00606*** (-5.09)
10th decile × RoO Strictness	-0.00312* (-2.02)
MFN Tariff	-0.00929* (-2.27)
2nd decile × MFN Tariff	0.00569 (1.36)
3rd decile × MFN Tariff	0.0137** (2.72)
4th decile × MFN Tariff	0.00922 (1.82)
5th decile × MFN Tariff	0.0102 (1.94)
6th decile × MFN Tariff	0.00972 (1.75)
7th decile × MFN Tariff	0.0162** (3.10)
8th decile × MFN Tariff	0.0228*** (3.58)
9th decile × MFN Tariff	0.0386*** (5.54)
10th decile × MFN Tariff	0.0397*** (4.60)
Constant	0.649*** (21.57)
Observations	103,917

t statistics in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.5: Estimated Marginal Effects of RoO and MFN tariffs.

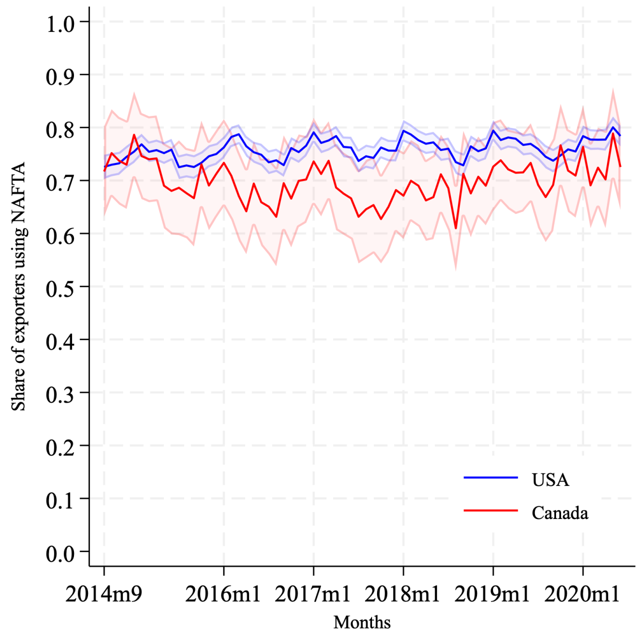


Figure D.1: Share of exporters using NAFTA by export destination.

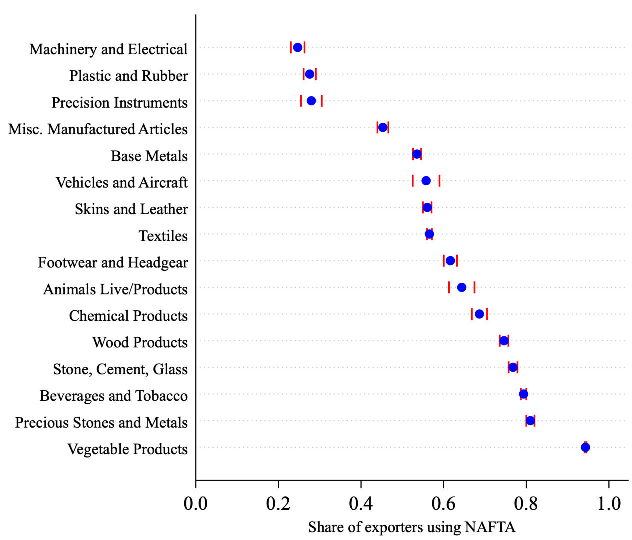


Figure D.2: Share of exporters using NAFTA by HS Section.

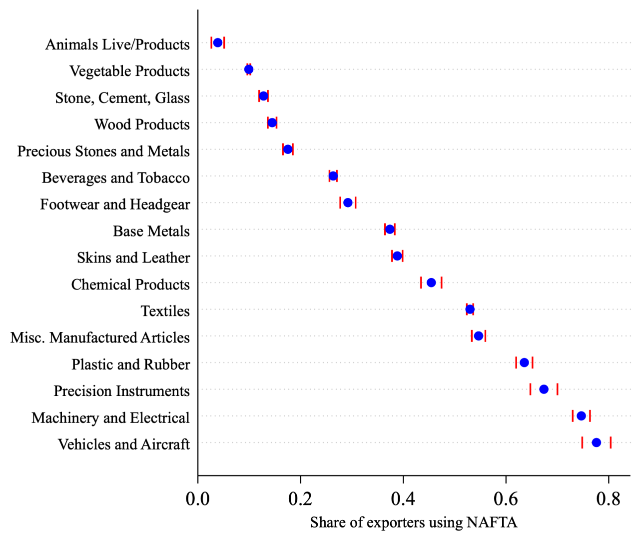


Figure D.3: Share of exporters sourcing outside of NAFTA by HS Section.

E Empirical vs. Model RoO Strictness

Let $\tilde{\lambda}$ denote RoO strictness as defined in Section 2, the share of a product's input value restricted to NAFTA suppliers. In our model, for a firm using NAFTA to export:

$$\tilde{\lambda} = \frac{\int_0^\lambda z_{si}(\nu) d^*\nu}{\int_0^\lambda z_{si}(\nu) d^*\nu + \int_\lambda^1 z_{si}(\nu) d\nu}$$

where λ is the model-based RoO strictness. We can rewrite the above as:

$$\tilde{\lambda} = \frac{\lambda \int_0^\infty z dG^*_{si}(z)}{\lambda \int_0^\infty z dG^*_{si}(z) + (1-\lambda) \int_0^\infty z dG_{si}(z)}$$

Under the properties of the Fréchet distribution we get:

$$\tilde{\lambda} = \frac{\lambda \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in N \cap J} T^h_{si} (d^h w^h_{si})^{-\theta} \right]^{-\frac{1}{\theta}}}{\lambda \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in N \cap J} T^h_{si} (d^h w^h_{si})^{-\theta} \right]^{-\frac{1}{\theta}} + (1-\lambda) \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in J} T^h_{si} (d^h w^h_{si})^{-\theta} \right]^{-\frac{1}{\theta}}}$$

For $\tilde{\lambda}$ to equal λ, i.e., for the empirical and model-based definitions of RoO strictness to coincide, we require $N \cap J = J$, which only holds if $J \subseteq N$. Intuitively, this occurs when a firm's sourcing strategy J consists solely of NAFTA countries in N. The definitions would also coincide if $J \nsubseteq N$ but all countries in $J \setminus N$ had zero sourcing potential $T^h_{si}(d^h w^h_{si})^{-\theta}$; however, such countries would not be part of a firm's sourcing strategy to begin with.

F Marginal Cost of Producing and Exporting

According to Equation (9), the marginal cost of producing and exporting is given by:

$$c_{si}(\phi, \kappa, \lambda, \tau, J) = \frac{1 + (1-\kappa)\tau}{\phi} \left(\int_0^{\kappa\lambda} z_{si}(\nu)^{1-\rho} d^*\nu + \int_{\kappa\lambda}^1 z_{si}(\nu)^{1-\rho} d\nu \right)^{1/(1-\rho)}$$

which we can rewrite as:

$$c_{si}(\phi,\kappa,\lambda,\tau,J) = \frac{1+(1-\kappa)\tau}{\phi}\left(\kappa\lambda\int_0^\infty z^{1-\rho}dG^*_{si}(z) + (1-\kappa\lambda)\int_0^\infty z^{1-\rho}dG_{si}(z)\right)^{1/(1-\rho)}$$

Taking into account that the distribution for prices depends on whether an input is restricted under RoO, $G^*_{si}(z)$, or is unrestricted, $G_{si}(z)$. Following Eaton and Kortum (2002):

$$\int_0^\infty z^{1-\rho}dG^*_{si}(z) = \Gamma\Big(\frac{\theta-1-\rho}{\theta}\Big)\left[\sum_{h\in N\cap J} T^h_{si}(d^h w^h_{si})^{-\theta}\right]^{\frac{-(1-\rho)}{\theta}}$$

and:

$$\int_0^\infty z^{1-\rho}dG_{si}(z) = \Gamma\Big(\frac{\theta-1-\rho}{\theta}\Big)\left[\sum_{h\in J} T^h_{si}(d^h w^h_{si})^{-\theta}\right]^{\frac{-(1-\rho)}{\theta}}$$

which allows us to rewrite marginal cost as:

$$c_{si}(\phi,\kappa,\lambda,\tau,J) = \frac{1}{\phi}\gamma^{-\frac{1}{\theta}}\Big[1+(1-\kappa)\tau\Big]\left[\kappa\lambda\Psi^{\frac{\rho-1}{\theta}} + (1-\kappa\lambda)\Phi^{\frac{\rho-1}{\theta}}\right]^{1/(1-\rho)}$$

with:

$$\gamma = \Gamma\Big(\frac{\theta-1-\rho}{\theta}\Big)^{-\frac{\theta}{1-\rho}}$$

$$\Psi = \left[\sum_{h\in N\cap J} T^h_{si}(d^h w^h_{si})^{-\theta}\right]^{\frac{-(1-\rho)}{\theta}}$$

$$\Phi = \left[\sum_{h\in J} T^h_{si}(d^h w^h_{si})^{-\theta}\right]^{\frac{-(1-\rho)}{\theta}}$$

G Purchases of Intermediate Inputs

This section shows how firm ϕ's purchases of inputs from country j can be expressed as a share of its operating profits, as stated in Equation (20). Using (13) we can write operating profits as:

$$\begin{aligned} \pi^o_{si}(\phi, \kappa^*, \lambda, \tau) &= [p_{si}(\phi, \kappa, \lambda, \tau, J) - c_{si}(\phi, \kappa, \lambda, \tau, J)]q_{si}(\phi) \\ \Rightarrow c_{si}(\phi, \kappa, \lambda, \tau, J)q_{si}(\phi) &= (\sigma - 1)\pi^o_{si}(\phi, \kappa^*, \lambda, \tau) \end{aligned} \tag{30}$$

As firm ϕ is sourcing share x^j_{si} of its inputs from country j, the share of its marginal cost coming exclusively from its purchases from j will be then given by:

$$\begin{aligned} c^j_{si}(\phi, \kappa, \lambda, \tau, J) &= x^j_{si}(\phi, \kappa, \lambda, J)c_{si}(\phi, \kappa, \lambda, \tau, J) \\ \Rightarrow c_{si}(\phi, \kappa, \lambda, \tau, J) &= \frac{c^j_{si}(\phi, \kappa, \lambda, \tau, J)}{x^j_{si}(\phi, \kappa, \lambda, J)} \end{aligned}$$

And then rewrite Equation (30) as:

$$\begin{aligned} q_{si}(\phi)\frac{c^j_{si}(\phi, \kappa, \lambda, \tau, J)}{x^j_{si}(\phi, \kappa, \lambda, J)} &= (\sigma - 1)\pi^o_{si}(\phi, \kappa^*, \lambda, \tau) \\ \Rightarrow q_{si}(\phi)c^j_{si}(\phi, \kappa, \lambda, \tau, J) &= (\sigma - 1)x^j_{si}(\phi, \kappa, \lambda, J)\pi^o_{si}(\phi, \kappa^*, \lambda, \tau) \end{aligned}$$

Since c^j_{si} is the marginal cost exclusively coming from j, i.e. the value of inputs purchased from j for the production of one unit of the final good, and $q_{si}(\phi)$ represents the number of units sold, it follows that:

$$M^j_{si}(\phi) = (\sigma - 1)x^j_{si}(\phi, \kappa, \lambda)\pi^o_{si}(\phi, \kappa, \lambda, \tau)$$

H Firm profits are not Supermodular

This section shows that, in our model, firm profits are not necessarily supermodular in productivity and thus need not exhibit increasing differences in a firm's sourcing strategy. As a result, we cannot apply the dimensionality reduction used in Antras et al. (2017) following Jia (2008), and instead must compute firm profits for every possible sourcing strategy.

According to Topkis's Modularity Theorem, if the profit function in Equation (16) is supermodular in $(I^j_{si}(\phi), \phi)$ —where $I^j_{si}(\phi) = 1$ if firm ϕ sources inputs from j —then $I^*_{si}(\phi) = (I^1_{si}(\phi), \dots, I^J_{si}(\phi))$ is non-decreasing in ϕ. In other words, the cardinality of a firm's sourcing strategy would increase with its productivity. For profits to be supermodular, two conditions must hold:

1. Let $X = [0,1]^J$ and $Y = \mathbb{R}^+$, both of which are lattices, so that $X \times Y$ is also a lattice. The profit function $\pi_{si}(\phi)$ must exhibit increasing differences in $(I_{si}(\phi), \phi)$ if and only if it has increasing differences in $(I^j_{si}(\phi), \phi)$ for all $j \in J$, holding $I^k_{si}(\phi)$ fixed for all $k \neq j$.

2. $\pi_{si}(\phi)$ features increasing differences in $(I^j_{si}(\phi), I^k_{si}(\phi))$, given $I^h_{si}(\phi)$ for $h \neq j, k$.

Our proof proceeds by showing that the profit function does not necessarily exhibit increasing differences in $(I^j_{si}(\phi), \phi)$. Consequently, it need not be supermodular, which implies that the cardinality of a firm's sourcing strategy may not increase with productivity. Let $\phi_H > \phi_L$. For profits to have increasing differences in $(I^j_{si}(\phi), \phi)$, the following condition must hold:

$$\begin{aligned}\mathbb{E}\Big[\pi_{si}(1,\phi_H) - \kappa(\phi_H)w\zeta_{si} - \pi_{si}(0,\phi_H) + \kappa(\phi_H)w\zeta_{si}\Big] &\geq \mathbb{E}\Big[\pi_{si}(1,\phi_L) - \kappa(\phi_L)w\zeta_{si} - \pi_{si}(0,\phi_L) + \kappa(\phi_L)w\zeta_{si}\Big] \\ \Rightarrow \mathbb{E}\Big[\pi_{si}(1,\phi_H) - \pi_{si}(0,\phi_H)\Big] &\geq \mathbb{E}\Big[\pi_{si}(1,\phi_L) - \pi_{si}(0,\phi_L)\Big]\end{aligned} \tag{31}$$

Since $I^{j'}_{si} \geq I^j_{si} \Rightarrow I^{j'}_{si} = 1 \wedge I^j_{si} = 0$, we focus on this case and omit $I^{j'}_{si} = I^j_{si} = 0$, which is trivially satisfied. Throughout, we fix the sourcing decision for all other countries in the firm's strategy, i.e. $I^{k'}_{si} = I^k_{si}$ for all $k \neq j$. Using the objective function in Equation (16), each side of the inequality can be written as:

$$\phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa)\tau]^{1-\sigma}[\Lambda(1,\phi)-\Lambda(0,\phi)] - w\mathbb{E}\Big[f^j_{si}(\phi)\Big] - w\sum_{k\neq j} I^j_{si}\mathbb{E}\Big[f^j_{si}(\phi)\Big] + w\sum_{k\neq j} I^j_{si}\mathbb{E}\Big[f^j_{si}(\phi)\Big]$$

where:

$$\Lambda(I^j_{si},\phi) \equiv [\kappa\lambda\Psi_{si}(I^j_{si},\phi)^{(\rho-1)/\theta} + (1-\kappa\lambda)\Phi_{si}(I^j_{si},\phi)^{(\rho-1)/\theta}]^{\frac{1-\sigma}{1-\rho}}$$

which allows us to rewrite Inequality (31) as:

$$\phi_H^{\sigma-1}[\Lambda(1,\phi_H) - \Lambda(0,\phi_H)] - w\mathbb{E}\Big[f^j_{si}(\phi_H)\Big] \geq \phi_L^{\sigma-1}[\Lambda(1,\phi_L) - \Lambda(0,\phi_L)] - w\mathbb{E}\Big[f^j_{si}(\phi_L)\Big]$$

Since we assume that fixed costs of sourcing do not depend on firm size, expectations cancel and we get:

$$\phi_H^{\sigma-1}[\Lambda(1,\phi_H) - \Lambda(0,\phi_H)] \geq \phi_L^{\sigma-1}[\Lambda(1,\phi_L) - \Lambda(0,\phi_L)]$$

By assumption $\phi_H > \phi_L$ and $\sigma > 1$, so for the above to necessarily hold true it needs to be the case that:

$$\Lambda(1, \phi_H) - \Lambda(0, \phi_H) \geq \Lambda(1, \phi_L) - \Lambda(0, \phi_L) \tag{32}$$

Since $\sigma > \rho$, Inequality (32) holds if $\phi_L \leq \phi_H \Rightarrow \Lambda(\phi_L) \leq \Lambda(\phi_H)$. Assuming $\phi_L \leq \phi_H$, the sourcing strategies $J(\phi_H)$ and $J(\phi_L)$ must satisfy that for the high-productivity firm:

$$\begin{aligned}
&\phi_H^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa(\phi_H))\tau]^{1-\sigma}\Lambda(J^*(\phi_H)) - w\sum_{j\in J^*(\phi_H)} I_{si}^j \mathbb{E}\Big[f_{si}^j(\phi_H)\Big] - \kappa(\phi_H)w\zeta_{si} \\
\geq\ &\phi_H^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa(\phi_L))\tau]^{1-\sigma}\Lambda(J^*(\phi_L)) - w\sum_{j\in J^*(\phi_L)} I_{si}^j \mathbb{E}\Big[f_{si}^j(\phi_L)\Big] - \kappa(\phi_L)w\zeta_{si}
\end{aligned}$$

and for the low-productivity firm:

$$\begin{aligned}
&\phi_L^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa(\phi_L))\tau]^{1-\sigma}\Lambda(J^*(\phi_L)) - w\sum_{j\in J^*(\phi_L)} I_{si}^j \mathbb{E}\Big[f_{si}^j(\phi_L)\Big] - \kappa(\phi_L)w\zeta_{si} \\
\geq\ &\phi_L^{\sigma-1}\gamma^{(\sigma-1)/\theta}B_{si}[1+(1-\kappa(\phi_H))\tau]^{1-\sigma}\Lambda(J^*(\phi_H)) - w\sum_{j\in J^*(\phi_H)} I_{si}^j \mathbb{E}\Big[f_{si}^j(\phi_L)\Big] - \kappa(\phi_H)w\zeta_{si}
\end{aligned}$$

The above follows from the fact that the high-productivity firm ϕ_H earns higher profits with $J(\phi_H)$, and similarly the low-productivity firm ϕ_L with $J(\phi_L)$. Adding these inequalities and noting that fixed sourcing costs are independent of productivity yields:

$$[\phi_H^{\sigma-1} - \phi_L^{\sigma-1}]\Big([1+(1-\kappa(\phi_H))\tau]^{1-\sigma}\Lambda(J^*(\phi_H)) - [1+(1-\kappa(\phi_L))\tau]^{1-\sigma}\Lambda(J^*(\phi_L))\Big) \geq 0$$

Since $\phi_L \leq \phi_H$ and $\sigma > 1$, this implies that:

$$[1+(1-\kappa(\phi_H))\tau]^{1-\sigma}\Lambda(J^*(\phi_H)) \geq [1+(1-\kappa(\phi_L))\tau]^{1-\sigma}\Lambda(J^*(\phi_L))$$

For $\Lambda(\phi_H) \geq \Lambda(\phi_L)$ to necessarily be the case, it has to be true that:

$$\begin{aligned}
&[1+(1-\kappa(\phi_H))\tau]^{1-\sigma} \leq [1+(1-\kappa(\phi_L))\tau]^{1-\sigma} \\
&\Rightarrow (1-\kappa(\phi_H))\tau \geq (1-\kappa(\phi_L))\tau \\
&\Rightarrow \kappa(\phi_H) \leq \kappa(\phi_L)
\end{aligned}$$

That is, whenever a high-productivity firm uses NAFTA to export, $\kappa(\phi_H) = 1$, any lower-productivity firm would also have to use it, $\kappa(\phi_L) = 1$. In our model, this need not hold: fixed costs of using NAFTA can lead to a medium-sized firm exporting under NAFTA while a smaller, less-productive firm does not. Since $\phi_L \leq \phi_H \nRightarrow \Lambda(\phi_L) \leq \Lambda(\phi_H)$, profits in our model do not necessarily exhibit increasing differences in $(I^j_{si}(\phi), \phi)$ and are therefore not necessarily supermodular in productivity. As a result, we cannot apply Topkis's Modularity Theorem to assert that the cardinality of a firm's sourcing strategy increases with productivity, and must instead brute-force the optimization by computing profits for each possible sourcing strategy.

Intuitively, this arises because the use of NAFTA and the presence of RoO introduce additional non-linearities into the model. For instance, a low-productivity firm may be unable to pay the fixed cost of using NAFTA and thus choose to source from non-NAFTA countries. A more productive firm might be able to pay this cost, opt for NAFTA, and as a result source exclusively from member countries. In this case, the set of countries available to the lower-productivity firm is actually larger than that of the higher-productivity firm.

I Full Results for Sourcing Potentials

For some industries in foreign countries, Mexican firms never import certain inputs, meaning the input shares for these product-region combinations are zero. If we were to drop these zero-share observations when regressing the log difference of the input share for a foreign region relative to Mexico against industry-region fixed effects, we would bias the estimated sourcing potentials upward. For example, suppose that in China, Industry A consists of Inputs 1-100, but Mexico imports only Input 100 from China, and this happens to be a product Mexico barely sources domestically. The resulting relative input share for China in Input 100 would be large. If Inputs 1-99 are excluded from the estimation, Input 100 becomes the sole observation, overstating China's sourcing potential for Industry A even though it represents only a small fraction of the industry's inputs. To prevent such bias, we retain all observations and assign an input share of 0.01 to any input-region combination that is not sourced from the foreign region.

Table I.1 reports the full set of estimated sourcing potentials for foreign countries, expressed relative to Mexico's sourcing potential (normalized to 1). For US-CA, the lowest value is in *Preparations of meat, or fish, or crustaceans* (0.08) and the highest in *Plastics and articles thereof* (6.60). For China, the minimum is also in *Preparations of meat, or fish, or crustaceans* (0.02), while the maximum is in *Glass and glassware* (1.71). For Europe, the lowest value is in *Meat* (0.02) and the highest in *Ceramic products* (0.71). For the Rest of the World, the highest sourcing potentials are in *Nuclear reactors, boilers, machinery and mechanical appliances* and *Electrical machinery and equipment and parts thereof*, while several sectors share the minimum value. Considering the total sum of estimated sourcing potentials, *Preparations of meat, or fish, or crustaceans* ranks lowest (0.13) and *Plastics and articles thereof* highest (8.26).

HS Chapter	Description	Sector	HS Section	SP US-CA	SP China	SP Europe	SP ROW	Total SP
02	Meat	1	1	0.15	0.03	0.02	0.01	0.21
03	Fish and crustaceans	1	1	0.38	0.18	0.06	0.06	0.68
04	Dairy Produce, Eggs, Natural Honey	1	1	0.87	0.26	0.14	0.02	1.29
06	Live trees and other plants	2	1	0.89	0.53	0.63	0.02	2.07
07	Edible vegetables	2	1	0.25	0.10	0.07	0.04	0.47
08	Edible fruits and nuts	2	1	0.22	0.09	0.05	0.03	0.40
16	Preparations of meat, or fish, or crustaceans	4	1	0.08	0.02	0.02	0.01	0.13
17	Sugars and sugar confectionery	4	1	1.58	0.61	0.38	0.02	2.58
18	Cocoa and cocoa preparations	4	1	2.21	0.87	0.56	0.01	3.65
19	Preparations of cereals, flour, starch or milk	4	1	0.95	0.34	0.20	0.03	1.52
20	Preparations of vegetables, fruit, or nuts	4	1	0.22	0.09	0.05	0.03	0.39
21	Mischellaneous edible preparations	4	1	0.64	0.22	0.12	0.03	1.00
32	Tanning or dyeing extracts	6	2	3.23	0.51	0.29	0.05	4.08
33	Essential oils and resionoids, perfumery, cosmetics	6	2	1.40	0.38	0.17	0.04	2.00
36	Explosibes, pyrothehnic products	6	2	0.53	0.18	0.08	0.03	0.82
39	Plastics and articles thereof	7	2	6.60	1.08	0.57	0.01	8.26
40	Rubber and articles thereof	7	2	2.39	0.62	0.26	0.02	3.30
42	Articles of leather	8	3	1.84	0.61	0.37	0.02	2.84
43	Furskins and artificial fur	8	3	0.96	0.40	0.14	0.03	1.53
44	Wood and articles of wood	9	3	1.00	0.22	0.12	0.03	1.38
46	Manufactures of straw, esparto or other plaiting materials	9	3	0.80	0.27	0.12	0.03	1.22
56	Wadding, felt and nonwovens, special yarns, ropes	11	3	3.39	1.04	0.43	0.02	4.88
61	Articles of appareal and clothing accessories, knitted	11	3	2.20	0.91	0.22	0.04	3.37
62	Articles of appareal and clothing accessories, not knitted	11	3	2.20	0.92	0.22	0.04	3.38
63	Other made up textile articles	11	3	3.00	1.11	0.39	0.01	4.52
64	Footwear, gaiters and the like	12	3	1.77	0.59	0.41	0.01	2.78
65	Headgear and parts thereof	12	3	2.12	0.76	0.23	0.04	3.15
66	Umbrellas, walking sticks, whips	12	3	1.44	0.29	0.14	0.04	1.91
67	Prepared feathers and down articles	12	3	1.44	0.29	0.14	0.04	1.91
69	Ceramic products	13	4	2.33	1.53	0.71	0.02	4.58
70	Glass and glassware	13	4	4.86	1.71	0.65	0.01	7.23
71	Precious stones, precious metals	14	4	0.58	0.28	0.14	0.05	1.04
73	Articles of iron or steel	15	4	0.89	0.21	0.10	0.04	1.24
76	Aluminum and articles thereof	15	4	0.74	0.14	0.07	0.05	1.00
83	Mischellaneous articles of base metal	15	4	0.87	0.22	0.10	0.04	1.24
84	Nuclear reactors, boilers, machinery and mechanical appliances	16	5	0.60	0.19	0.07	0.07	0.92
85	Electrical machinery and equipment and parts thereof	16	5	1.02	0.33	0.13	0.07	1.55
87	Vehicles other than railway or tramway	17	5	0.83	0.30	0.12	0.06	1.31
90	Optical, photographic, precision, medical apparatus	18	5	2.23	0.56	0.27	0.06	3.12
91	Clocks and watches and parts thereof	18	5	0.87	0.81	0.48	0.05	2.22
92	Musical instruments	18	5	1.40	0.29	0.14	0.04	1.86
94	Furniture, bedding, mattresses, cushions, lamps	20	6	1.68	0.49	0.19	0.04	2.41
95	Toys, games and sports requisites	20	6	3.18	0.66	0.38	0.03	4.25
96	Miscellaneous manufactured articles	20	6	1.57	0.33	0.17	0.04	2.11
Average				1.55	0.49	0.24	0.03	2.31
Minimum				0.08	0.02	0.02	0.01	0.13
Maximum				6.60	1.71	0.71	0.07	8.26

Table I.1: Industry-level Sourcing Potentials by Foreign Country.

J Point-Estimates using SMM

Table J.1 presents the point estimates from our SMM estimation. The estimates for the location parameter of the log-normal distribution for fixed sourcing costs are based on our simplifying assumption for the shape parameter, $\delta_s^j = \sqrt{\log 3}$, which implies that the variance of fixed sourcing costs is given by $\mathbb{V}(f_{si}^j) = \exp(2\mu_{si}^j + \log 3)$.

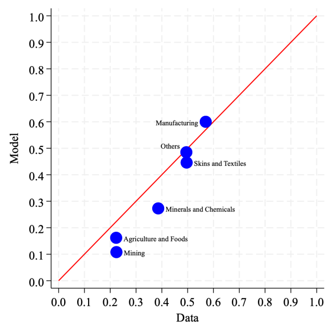
Sector	ζ_s	f_s^{CHN}	f_s^{EUR}	$f_s^{US\text{–}CA}$	f_s^{ROW}	B_s
Agriculture and Foods	0.011	0.143	0.149	0.147	0.035	0.048
Minerals and Chemicals	0.002	0.072	0.026	0.149	0.003	0.01
Skins and Textiles	0.006	0.053	0.031	0.148	0.003	0.016
Mining	0.00	0.021	0.036	0.04	0.003	0.002
Manufactures	0.022	0.023	0.019	0.097	0.006	0.027
Others	0.014	0.064	0.033	0.146	0.005	0.019
Average	0.009	0.063	0.049	0.121	0.009	0.021

Table J.1: Point Estimates for Fixed Costs and US Market Demand.

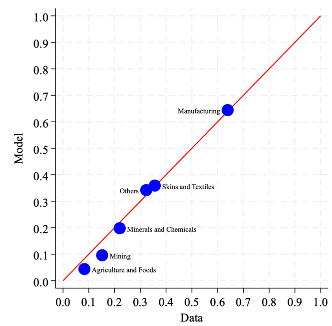
K Further Details on the Fit of the Model

Figure K.1 shows the model's predictions in terms of the extensive margin of sourcing: the share of firms, for each sector, that source inputs from either NAFTA, China, Europe, or the Rest of the World.

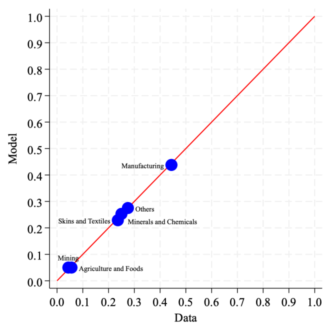
Figure K.2 shows the model's fit for the share of inputs sourced from each foreign region. These moments are linked to those in Figure K.1, since a firm not sourcing from a region implies an input share of zero, while sourcing from a region sets shares according to the estimated sourcing potentials in Section 5.2. The model matches entry rates into foreign regions well across all sectors —these are targeted moments in our structural estimation. However, the fit for regional input shares is mixed: we overpredict shares for the US-CA region and the Rest of the World, underpredict those for China, and match Europe's shares well across all sectors.



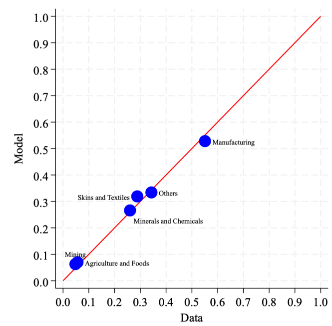
(a) US-CA



(b) China

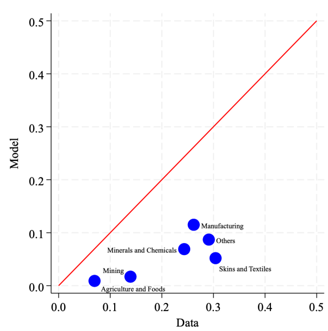


(c) Europe

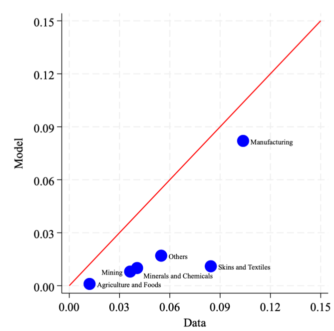


(d) Rest of the World

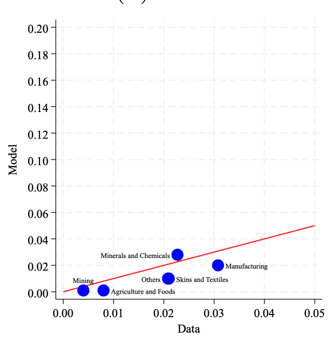
Figure K.1: Sectoral fit of the model in terms of the extensive margin.



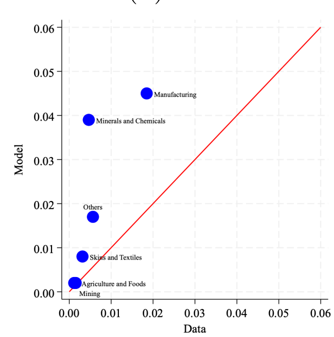
(a) US-CA



(b) China



(c) Europe



(d) Rest of the World

Figure K.2: Sectoral fit of the model in terms of input shares.